Cell-mean fluctuations in an arithmetic convolution field: exact Rademacher-null variances and a permutation control

Youngrok Lee
Independent researcher, Republic of Korea

Abstract

Let $Z(N) = C(N)/\sqrt{V(N)}$ with $C(N) = \sum_{n<N} \Lambda(n)\mu(N-n)$, and partition a band of even N into cells indexed by how many of $3, 5, 7, 11, 13$ divide N. Whether a cell mean departs from the band mean needs a reference scale, and the count-based one misses it by a factor growing with the band.

Under the null of independent Rademacher signs for the $\mu(v)$ on their support, the variance of any cell contrast is a doubly centred quadratic form in the covariance kernel and reduces to three computable terms built from finite correlations and weighted inner products: the exact null scale is written down, not estimated. For a uniformly random same-size cell, the permutation average of the sign-null variance equals the finite-population formula exactly; the divisibility cells do not. Measured over eight octaves to $1.6 \cdot 10^7$, the exact null standard deviation shrinks by 1.21 where a count predicts 11.3, and none of ten sampled label permutations preserving every cell size reproduced the floor; their collapse factors range from 3.8 to 105.

For its size we offer a descriptor: the excess D_c of the shift's singular series over same-cell pairs, whose main term is rational with no fitted parameter. For a second, differently shaped partition, the exhaustive measurement at the three scored depths falls inside the interval computed from the residue main term. The fourth depth is an algebraic consistency check, not a scored interval. At this band, D_c tracks the null floor with correlation 0.9805 across six cells; this post hoc association was not transferred. Every registered rule is reported with its verdict, refuted ones included. No Gaussian or limiting distributional law for Z and no Goldbach consequence is asserted, and the decay of the effect with N is not determined here.

1 Introduction

1.1 The setting

Let N range over the even integers of a band, let

$$C(N) = \sum_{n<N} \Lambda(n)\mu(N-n), \qquad V(N) = \sum_{v<N} \mu^2(v)\Lambda(N-v)^2, \qquad Z(N) = \frac{C(N)}{\sqrt{V(N)}}, \tag{1}$$

so that Z has second moment 1 under the null in which the $\mu(v)$ on their own support are replaced by independent signs; the exact asymptotic $V(N) \sim \prod_{q \nmid N}(1 - 1/(q(q-1)))\, N \log N$ is Proposition 21 in Appendix A and is quoted for scale only: no computation below uses it, every one dividing by $V(N)$ as computed. The heuristic of Observation 5 does read the scale $N \log N$ off it, and nothing else here does. The field (1) is the scalar to which the Huang–Li reduction of binary Goldbach reduces: their Theorem 1, with Corollary 1 drawing the Goldbach conclusion from it, turns on the equidistribution of $\Lambda(n)\mu(N-n)$ in arithmetic progressions [1, Thm. 1, Cor. 1]. That is imported, not derived here, and used for nothing below, since §1.3 refuses the Goldbach consequence outright. Partition the band into *cells* indexed by *depth* d, the number of $3, 5, 7, 11, 13$ dividing N. The question is whether $m_c - \overline{m}$, the difference between the cell mean of Z and the band mean, is larger than the sign null admits. *Chance* is not the word for it and is not used below: that null replaces μ on its own support by independent signs and so discards its multiplicativity, and Note 11 reports an ensemble that keeps the multiplicativity as the wider of the two, with no transfer between them established here. Every standardised figure below is against the sign null.

Answering it requires a reference scale for $m_c - \overline{m}$. The reflex is $\mathrm{sd}(Z)/\sqrt{n_c}$ with n_c the cell count. That the reflex fails for correlated observations is standard — the variance of a weighted mean is $w^\top \Sigma w$, and the ratio to the independent formula is the design effect [6]. What is not standard, and is what this paper supplies, is that for the field (1) the covariance Σ is *exactly computable* under a sign null; that cells cut by divisibility align with it; that for a uniformly random same-size cell, the permutation average of the sign-null variance equals the finite-population formula exactly; and that the size of the effect is predicted by an excess of the shift's singular series over same-cell pairs.

1.2 Results

- Lemma 2 gives $\mathrm{Var}(m_c - \overline{m})$ exactly under the sign null, as three explicit terms built from finite correlations and weighted inner products.
- Observation 5 takes up its decay: over the eight measured octaves the shallow cells are consistent with $(\log N)^{-1/2}$ and not with $n_c^{-1/2}$. The first is a heuristic read against a finite range, not a proved rate; what is established over the measured range is that count-based scaling does not describe the arithmetic-cell floor.
- Lemma 8 is the control that separates a property of the cell *sizes* from a property of the correspondence between cells and arithmetic.
- Measurement 10 runs it, and finds that neither the cell effect nor the arithmetic-cell floor is reproduced under the placebo; the floor itself collapses, so it is a second measurement of the same correspondence.
- Proposition 17 shows the quantity offered for the floor's size is asymptotically scale-free: as the band grows it converges to a constant fixed by the cell's residue classes, so its one-step exponent tends to zero wherever that constant is nonzero.
- Measurement 18 computes that quantity's main term in exact rational arithmetic and finds it reproduces the measured profile at all six depths with no fitted parameter — which both supplies the hypothesis the proposition needs (that the main term does not vanish) and turns the proposition from a statement that the size does not drift into one that names it.
- Measurement 19 carries the same computation to a *second* cell shape, computing its profile from the residues alone and then evaluating the finite pair average exhaustively. The computed intervals contain the exact values where the rules could score them, and the small correction it carries decomposes with no free parameter into a divisor deficit and a covariance — the second of which is exactly zero at the only cell where the first was ever calibrated.

1.3 What is not claimed

- **No decay exponent for the effect is quoted, and the shallow depths are not uniformly quiet.** That the amplitude at the three shallowest depths clears nothing at any measured scale cannot be read off two octaves, and the two whose per-depth amplitudes were printed are not the quietest two: one is the smallest of the eight and the other the third smallest. The run behind Measurement 9 prints all eight, and at depth 1 the amplitude runs

$$+2.71,\ +2.18,\ +1.96,\ +1.72,\ +1.34,\ +1.11,\ +0.87,\ +0.69$$

from $(6.25 \cdot 10^4, 1.25 \cdot 10^5]$ up to $(8 \cdot 10^6, 1.6 \cdot 10^7]$ — the value at the smallest octave exceeding the -1.55 that depth 3 carries at the largest, where the paper calls the deep cells the signal. So no threshold separating "clears" from "does not" is asserted at the shallow depths. What is asserted is the shape: at each of depths 0, 1 and 2 the amplitude falls monotonically across all eight octaves, and no exponent is fitted to that fall. Whether the exponents differ by depth is not decided by the accessible range.

- **No Gaussian or limiting distributional law for Z is asserted.**
- **No consequence for the Goldbach problem.** Nothing measured here bounds $C(N)$ in either direction. No sweep over parameterisations is carried out, and none is claimed.
- **That D_c predicts the floor's size is measured, not derived.** The exact doubly centred singular-series analogue of the variance is B_c, defined in §5; D_c itself is the two-term descriptor in (7). Measurement 15 shows that D_c and B_c are close in the measured band because the cross average of the singular series is close to its band average. A constant ratio between the variance and B_c would follow if the two kernels differed across the band by a factor the centring leaves alone. No such statement is proved here. What is measured is the ratio: its full width is 25.4% of its mean, or 19.8% once se_c^2 is matched to D_c's convention term by term — matched that way the row is still monotone from depth 1 on, with a last step nine times smaller than the printed one — while D_c moves by a factor 68, and the correlation 0.9805 carries an exact enumeration over all 720 relabellings as its control. That is one octave read across depths. Read across scale instead, the same ratio drifts by about half as much again (Measurement 15), so what is measured is a depth profile at a fixed scale and D_c is not all of what fixes the size. (6) is a difference of three means and not of two, and Measurement 15 records what dropping the third would cost.
- **Observation 5 is offered for its form, not its constant.** Its heuristic is quantitatively loose by factors running from 2.49 at depth 0 to 6.20 at depth 5 — the 2.5 is the shallowest cell, not the range, whose lower end is the 2.45 at depth 1; the exponent is confirmed directly instead.
- **The statements here are not equally checked.** The derivation of Observation 5 and the statement of Lemma 8 have had less scrutiny than the rest, and Proposition 17 has had no reading by a subject expert. What was checked by what is set out in Supplement S1 (Appendix B).

2 The floor, in closed form

The identity this section rests on is not arithmetic. It is the statement that a cell contrast is a linear functional of the observation vector, so its variance is the quadratic form $w^\top \Sigma w$ in whatever covariance Σ the observations carry. We state that first, in the generality in which it is true, and specialise afterwards.

Lemma 1 (contrast variance under an arbitrary kernel)**.** *Let $X = (X_N)_{N\in a}$ be any square-integrable random vector indexed by a finite set a of size n, with covariance $\Sigma_{NN'} = \mathrm{Cov}(X_N, X_{N'})$. For $c \subseteq a$ of size $n_c \geq 1$ let $m_c = n_c^{-1}\sum_{N\in c} X_N$ and $\overline{m} = m_a$. Writing $Q_{cd} = \sum_{N\in c}\sum_{N'\in d}\Sigma_{NN'}$,*

$$\mathrm{Var}\big(m_c - \overline{m}\big) \;=\; \frac{Q_{cc}}{n_c^2} \;-\; \frac{2\,Q_{ca}}{n_c\,n} \;+\; \frac{Q_{aa}}{n^2}. \tag{2}$$

Proof. $m_c - \overline{m} = w^\top X$ with $w_N = n_c^{-1}\mathbf{1}_{N\in c} - n^{-1}$, so the variance is $w^\top \Sigma w$. Expanding gives the four products of the two terms of w against themselves; they collapse to three sums because the two cross terms are equal, $Q_{ac} = Q_{ca}$ by symmetry of Σ. □

Nothing above uses independence, stationarity, or a distribution. What the null supplies is not the identity but the kernel: it makes Σ diagonal in a basis where it can be summed, and hence makes Q_{cd} computable rather than merely defined. That is the content of the next lemma, and it is where the arithmetic enters.

Lemma 2 (exact cell moments)**.** *Let c be any set of n_c of the even N of a band of size n — the proof uses nothing beyond $c \subseteq a$, and in this paper the cells do partition the band — let m_c and $\overline{m}$ be the means of Z over c and over the band, and set*

$$u_c(v) \;=\; \sum_{N\in c} \frac{\Lambda(N-v)}{\sqrt{V(N)}}, \qquad Q_{cd} \;=\; \sum_{1\le v<N_{\max}} \mu^2(v)\, u_c(v)\, u_d(v),$$

where $N_{\max}$ is the largest element of the band and $\Lambda(k) = 0$ for $k \le 0$, so that the inner sum runs over the N of c that exceed v and the outer one over the squarefree v below the band's top. Then, when the $\mu(v)$ on the surviving support are replaced by independent signs,

$$\operatorname{Var}\bigl(m_c - \overline{m}\bigr) \;=\; \frac{Q_{cc}}{n_c^2} \;-\; \frac{2\,Q_{ca}}{n_c\, n} \;+\; \frac{Q_{aa}}{n^2}, \tag{3}$$

where a denotes the whole band. The quantities u_c are finite correlations of explicit sequences, and each Q_{cd} is then an explicit weighted inner product; the identity is exact and no simulation is involved. Its numerical evaluation is a separate matter — these finite operations are carried out in double precision (Appendix B), so each printed floor is this exact formula evaluated to floating-point accuracy and not an exact rational.

Proof. Under independent signs the covariance carries μ^4, and $\mu^4 = \mu^2$ because μ takes only 0 and ± 1; so $E[\mu^2(v)\varepsilon(v)\cdot\mu^2(v')\varepsilon(v')] = \mu^2(v)\delta_{vv'}$ and $\operatorname{Cov}(n_c m_c, n_d m_d) = Q_{cd}$ exactly, which is the Q_{cd} of Lemma 1 for this Σ. That lemma then gives the three terms. □

For later use, write

$$\mathrm{se}_c := \sqrt{\operatorname{Var}_\varepsilon(m_c - \overline{m})}, \qquad z_c := \frac{m_c - \overline{m}}{\mathrm{se}_c}.$$

Throughout, the *null standard deviation* of a cell contrast means se_c: the exact standard deviation of $m_c - \overline{m}$ under the Rademacher null, at the observed cell sizes and the observed support of μ^2. It is a scale, not a tail bound. The lemma fixes the scale exactly, but no Gaussian, limiting, or other calibrated tail law is invoked for $m_c - \overline{m}$; no probability is attached below to a multiple of se_c merely because it is such a multiple.

Note 3 (all three terms are needed). The substitution $u_c(v) \approx n_c/\sqrt{V}$, which gives the right scale for Q_{cc}/n_c^2, is not specific to c; applied to all three terms it returns the same value S for each and hence $S - 2S + S = 0$. So it cannot be used to estimate the variance of the difference. Exactly, the variance is smaller than Q_{cc}/n_c^2 by a factor that is constant *in* N rather than noise — constant for a fixed cell, not across cells: at one N it reads 0.113, 0.016, 0.137, 0.289, 0.427, 0.552 from depth 0 to depth 5. That row is not monotone either — its minimum 0.016 sits at depth 1, a seventh of the value at depth 0 — so the spread across cells is a factor 34 and not the factor 5 that reading the two ends alone would give. Across the eight octaves it holds to four digits at depths 0 and 1 and to three at depths 2 and 3, drifts by 2.2% at depth 4 and by 30% at depth 5, whose cell grows from 2 members at the lowest octave to 266 at the highest: the constancy is a statement about the shallow cells, where there are enough N for it to be one.

Measurement 4 (the closed form, checked against simulation)**.** The lemma does not need simulation, but simulation is the only thing that would catch an error in the formula, and every normalised cell mean below is divided by it. Run in the band $(10^5, 2\cdot 10^5]$ with 2000 draws:

depth	0	1	2	3	4	5
n_c	19181	21097	8183	1421	115	3
closed form	0.140080	0.052724	0.167579	0.273630	0.378073	0.639330
MC/closed	0.9920	0.9993	0.9918	1.0044	1.0157	1.0320

— worst deviation 0.0320, at the depth whose cell holds three elements, and the six ratios straddle 1 three and three.

The six ratios are estimated from the *same* draws, so their deviations are strongly correlated and their common sign would be one observation, not six. At 2000 draws the Monte-Carlo precision is $1/\sqrt{2\cdot 1999} = 0.015815$, and the six deviations stand to it in 0.507, 0.041, 0.519, 0.275, 0.992, 2.025. Against that bar only the cell of three elements clears, and clearing it is not agreement.

That bar is a convention and not the precision. $1/\sqrt{2(n-1)}$ is the Gaussian value of a delta-method approximation to the relative standard error of a sample standard deviation; it is not exact for any distribution, and the sampling variance of a sample standard deviation has no closed form here. What is exact is one level down: for the sample *variance* of n independent draws the sampling variance is $n^{-1}(\mu_4 - \frac{n-3}{n-1}\sigma^4)$, and μ_4 is available, since each contrast is a weighted Rademacher sum $\sum_v a_v\varepsilon_v$ with $\mathbb{E}S^4 = 3(\sum a^2)^2 - 2\sum a^4$.

Two things follow and only the second is used. The kurtosis is $3 - 2\sum a^4/(\sum a^2)^2$, below the Gaussian 3 at every cell. So the approximation the bar comes from is, to leading order, an *over*estimate, and it is one that varies from cell to cell since the weights do. The direction is what matters for the sentence above: a cell-specific bar is tighter, so "only the cell of three clears" is a statement about the common bar and not about the cells. What the common bar supports is the weaker reading, that the deviations are small on the scale the draw count alone fixes. The cell-wise fourth moments are not computed here and no exact interval is claimed. That is one deviation of that size among six correlated estimates and it stays inside the registered cap of 0.05, so it refutes nothing; what it shows is that the resolution and the disagreement sit at the same cell. Of the remaining five, four sit at 0.52 of the bar or less, and depth 4 sits at 0.992 of it — short by 0.8 per cent of the bar, at the precision and not below it. At four cells the table resolves nothing, at one it sits at the bar, and at a number of draws small enough for the precision to exceed every deviation in the table it would resolve nothing at all. Two of the four rules registered in the result file score a 60-draw quotation this paper does not make; the ratios of this paragraph are checked in a block marked post hoc there, and were not registered in advance.

The structural constant of Note 3 is visible directly: at the top octave of $1.6\cdot 10^7$, $\mathrm{Var}/(Q_{cc}/n_c^2)$ is 0.113119, 0.016302, 0.288567, 0.551516 at depths $0, 1, 3, 5$, and at a quarter of that N the three shallow ratios agree to five decimals — 0.113118, 0.016303, 0.288571, the largest gap being $4\cdot 10^{-6}$. Depth 5 does not: it reads 0.557654 there against 0.551516 at the top, differing in the third digit, and it is much the smallest cell — 266 values of N in the top octave against 1 687 911 at depth 1 (`lab_cellmom_montecarlo.py` for the table, `lab_cell_floor.py` for this paragraph).

3 The floor's uncertainty does not fall like a count

Observation 5 (coherent sums)**.** The null standard deviation of Lemma 2 does *not* decay like $n_c^{-1/2}$. On average over the v that carry weight, about $n_c/\log N$ of the terms of $u_c(v)$ are nonzero, each of size $\log N/\sqrt{V}$, so $u_c(v) \approx n_c/\sqrt{V}$ and

$$\frac{Q_{cc}}{n_c^2} \approx \frac{\sum_v \mu^2(v)}{V} \asymp \frac{1}{\log N},$$

independently of n_c. *At a fixed v this is false*: the band is even, so an even v leaves $N-v$ even and $\Lambda(N-v)$ supported on powers of 2 alone, and the cell's own primes impose further local obstructions. What the display uses is the sum over v, where those obstructions are the local densities being averaged; the heuristic is offered at that level and not at a single v. By Note 3 the variance itself is a fixed fraction of this. So the standard error falls like $(\log N)^{-1/2}$ — at the shallow cells, which is where Note 3 says that fraction is constant. It does not extend to the deepest: the same note records a thirty per cent drift of the fraction at depth 5, and the exponent fitted there — 0.107306, in the ungated rows of `lab_cell_floor.txt` rather than in Measurement 6, which prints the three gated depths — is the largest of the six. That drift is not the whole of the break. Separately from it, and not confined to the smallest cell, Q_{cc}/n_c^2 does not follow $1/\log N$ at any depth. Over the eight octaves $Q_{cc}n_c^{-2}\log N$ would be constant if the form held. It falls, and its value at the lowest octave stands to its value at the highest in 1.031 at each of depths 0 to 3, 1.049 at depth 4 and 1.621 at depth 5. **So the form is loose at every depth, and by about the same amount at the shallow five** — but not at the deepest, where the drift is larger by an order of magnitude: $\log 1.621/\log 1.031$ is 15.8. The uniformity claimed here is over depths 0 to 4 and no further, and depth 5 is the single-class cell that the rest of this section treats separately. The drift is quoted as that ratio and not as a percentage because a percentage needs a denominator: the two obvious choices differ by the ratio itself, which is immaterial at 1.03 and is not at 1.62. For the same reason no value is quoted for the fitted exponent g of that quantity. Its *sign* is negative at all six depths under each of the three octave abscissae — the low endpoint, the midpoint, the high. Its *value* moves between those three by 0.003721 at every one of the six depths, the same number at all of them: the slope of $\log(Q_{cc}n_c^{-2})$ is untouched, since the abscissae differ by the constant $\log 2$, so the whole dependence is the shared $\log N$ factor and an additive term is what a constant swing looks like. Against $|g|$ that swing is most of the exponent where $|g|$ is small and four per cent at depth 5, which is why no value is quoted.

That drift is the measured exponent in another coordinate and not a second reading of it. Where the ratio $\mathrm{Var}/(Q_{cc}n_c^{-2})$ is constant in N — its spread across the eight octaves is 0 at depths 0 and 1 — one has $-2b$ for the slope of $\log(Q_{cc}n_c^{-2})$ in $\log N$, while g is that slope plus $1/\log N$; so $b = 1/(2\log N) - g/2$ identically, and the two sides cannot disagree. Nothing in this paragraph should be read as the drift confirming the exponent.

What the residual of that identity does read is the constancy of the ratio itself, since that is its only hypothesis, and it is there that the deepest cells separate. Half the negated slope is 0.039408 against the fitted $b = 0.039388$ at depth 0, a difference of $2 \cdot 10^{-5}$ where the ratio does not move at all; at depth 4 the difference is 0.0019 against a ratio spread of 0.0094, and at depth 5 it is 0.026 against a spread of 0.165. So the break at the deepest cell is the failure of the ratio to hold fixed. The diagonal $1/n_c$ of $Q_{cc}n_c^{-2}$ rises across the same two depths — its share is at most 0.42 per cent at depths 0 to 3 and 70.5 per cent at depth 5, at the lowest octave — and this measurement does not separate the two, so the diagonal is reported alongside the break and is not offered as its cause. The exponent is confirmed directly at all six depths and that confirmation, not the form, is what the paper rests on.

The heuristic is quantitatively loose and is offered for the *form* and not the constant: at $N = 1.6 \cdot 10^7$,

depth	0	1	2	3	4	5
Q_{cc}/n_c^2	0.123577	0.121208	0.146026	0.184159	0.235171	0.307074

against the heuristic's $\left(\sum_{v<N}\mu^2(v)\right)/V(N) = 0.049540$ (`lab_cell_floor.py`). The row is not monotone in the depth; its minimum is 0.121208 at depth 1.

Measurement 6 (the exponent, fitted)**.** The form is confirmed directly. Fitting se $\propto N^{-b}$ to the exact floor across eight octaves gives $b = 0.039451, 0.039671, 0.039388$ at depths $2, 1, 0$, against the 0.5 that a count would give and against 0.036038, the value of $1/(2\langle\log N\rangle)$ with $\langle\log N\rangle$ the mean of $\log N$ over the eight octave midpoints. Nothing turns on the third decimal: the point is that 0.039 is near 0.036 and nowhere near 0.5 (`lab_cell_floor.py`).

Note 7 (a count is not the null standard deviation). The consequence is not confined to this field. The fit runs over the eight octaves from $(6.25 \cdot 10^4, 1.25 \cdot 10^5]$ to $(8 \cdot 10^6, 1.6 \cdot 10^7]$, and the abscissa is the octave midpoint — which is what makes $1/(2\langle\log N\rangle) = 0.036038$ come out, with $\langle\log N\rangle = 13.8744$. The factor in N is therefore $1.2 \cdot 10^7/93750 = 128$. Over it, $n_c^{-1/2}$ says the null standard deviation shrinks by $\sqrt{128} = 11.3$; it shrinks by $128^{0.0395} = 1.21$, against the 1.19 that $(\log N)^{-1/2}$ predicts over the same factor. The result file prints the shrink itself only from depth 3 on; at the three gated depths it is 128^b from the exponents printed there, which is the one step taken here. **An interval built from a count is therefore, against the exact Rademacher-null scale, a factor $11.3/1.21 = 9.3$ too narrow at the top of this range relative to the bottom**, and in absolute terms the discrepancy is larger still: at the top octave the exact floor exceeds $\mathrm{sd}(Z)/\sqrt{n_c}$ by factors of 7.3 to 158, with $\mathrm{sd}(Z) = 0.924687$ over that octave — and 0.929384 over the whole fitted field, so the reading of sd barely moves the figures. The factor does *not* grow with the cell. What moves with the cell is the floor itself, which reads 0.1182, 0.0445, 0.1414, 0.2305, 0.3169, 0.4115 from depth 0 to depth 5: it is not monotone — depth 1 sits below depth 0, the same dip that Measurement 15 reads in D_c and in se_c — and it grows from depth 1 onward. That movement is measured and not predicted by $Q_{cc}/n_c^2 \asymp 1/\log N$: that relation is a statement about N, carries no dependence on the cell, and so says nothing about how the floor moves across depths at one N. Section 1.3 records the same thing from the other side — the heuristic behind it is loose by a factor that is 2.49 at depth 0 and rises to 6.20 at depth 5, with its minimum 2.45 at depth 1 rather than at either end. The factor also carries $\sqrt{n_c}$, and n_c runs 1 534 466, 1 687 911, 654 281, 114 017, 9 059, 266 across those depths, which is the larger movement. That row is not monotone either: its maximum is at depth 1, so the largest cell is the one where Measurement 15 reads the smallest D_c and the smallest floor. By depth the factor reads 158, 62, 124, 84, 33, 7 (`audit_cellfloor_sdz.py`). The cause is that $u_c(v)$ is a coherent sum, not a self-averaging one, and that the cell is correlated with the arithmetic it sums. Coherence alone does not give it. A cell drawn at random from the same band is coherent in the same sense — every N reads the same μ and Λ at the same v — but there $u_c(v) \approx (n_c/n)u_a(v)$ and the three terms of (3) cancel to leading order, so Lemma 12 returns exactly $n_c^{-1/2}$, up to the finite-population factor, for the average over the permutation. What survives the cancellation is the departure from that proportionality, and only a cell correlated with the arithmetic produces one.

4 The control, and what running it shows

Lemma 8 (the placebo key)**.** *Let the cells be indexed by a labelling $\ell(N)$, and let π be a uniformly random permutation of the even integers in the band. This is a randomisation* conditional on the observed cell counts*: the labels are not drawn independently, since the counts are held fixed, and the field Z is held fixed with them — the standard randomisation-inference setting [7]. Call a statistic* cell-indexed *when it is a function of the pair (ℓ, Z) — of which N carry which label, and of the field. Replacing ℓ by $\ell \circ \pi$ preserves every cell size and leaves the field Z pointwise unchanged, while destroying the correspondence between cells and arithmetic. Any cell-indexed statistic determined by the multiset $\{Z(N)\}$ together with the cell sizes — and so not by that correspondence — has its distribution unchanged by the replacement, and is in fact constant under it. The converse fails, and the lemma does not assert it: $T(\ell, Z) = Z(N_0)$ at a fixed N_0 of the band is cell-indexed and invariant, since it does not read ℓ at all, yet the multiset and the sizes* *do not determine it. What the control gives is the direction stated, used below in contrapositive. The determining data need not be the sizes alone: s^2/n_c, with $s^2 = n^{-1}\sum_N (Z(N) - \overline{m})^2$ the variance of the field over the band, is invariant under the replacement and reads the field through s^2 — and it is, up to a finite-population factor, the permutation null (5) that Lemma 12 will give. Not the permuted floor (4) itself: that is the sign-null expectation of this one, carrying $E_\varepsilon s^2 = 1 - Q_{aa}/n^2$ where this carries s^2; Note 13 reads both off the measured octave. Nor is the floor at a single π determined by the two invariants: (6) reads K on the subset actually drawn and K is not constant across the band. What the invariants fix is the average over π, which is what (4) is — and the control reports the ten-draw standard error of that average beside it.*

Conversely the replacement supplies a null. The law of a cell-indexed statistic under $\ell \circ \pi$ is its law under labels drawn independently of Z; and a statistic determined by the two invariants is constant under the replacement, so its null is a point mass. Hence a statistic whose true-labelling value that null does not reproduce is not determined by the two multisets, and what it reads is the pairing between them — the correspondence the replacement destroys. That is the whole of what is claimed: which statistics are uninformative, not that the permutation law is the right null for any other question.

Proof. The permutation acts on labels only; the multiset $\{Z(N) : N \text{ in the band}\}$ and the multiset of cell sizes are both invariant, and the joint distribution of $(\ell \circ \pi, Z)$ is that of a uniform fixed-count relabelling, with the random relabelling independent of Z. That observation gives both halves: the first reads it as the law the statistic must have if it sees only those two invariants, the second as the reference law against which the true labelling is read. For the second, a statistic determined by the two invariant multisets takes one and the same value at every π. □

Measurement 9 (the cell contrasts are not zero)**.** Against the floor of Lemma 2, computed from its closed form, the cell contrasts $m_c - \overline{m}$ of the field (1) are not all zero. Over the eight octaves from $(6.25 \cdot 10^4,\ 1.25 \cdot 10^5]$ to $(8 \cdot 10^6,\ 1.6 \cdot 10^7]$, $\max_c |z_c|$ reads

$$10.099,\ 10.954,\ 11.179,\ 12.968,\ 11.885,\ 11.026,\ 10.019,\ 9.064,$$

No multiple-comparison threshold is asserted for these. The finite Rademacher sign ensemble does define a joint law, but this paper does not calibrate the tail of $\max_c |z_c|$ under that law. The numbers are therefore reported only as a maximum over cells of standardised deviations against the closed-form null floor. The signal is carried by the deep cells: at the top octave depths $3, 4, 5$ sit at $z = -1.55, -4.46, -9.06$ while depths $0, 1, 2$ sit at $+0.04, +0.69, -0.06$ (`lab_cell_floor.py`).

Measurement 10 (the placebo destroys the effect, and the floor with it)**.** Run on the octave $(2 \cdot 10^6,\ 4 \cdot 10^6]$, with cell sizes preserved exactly and the floor recomputed from scratch for each permutation:

depth	0	1	2	3	4	5
n_c	383617	421978	163568	28507	2263	67
z_c (true)	+0.1069	+1.1133	−0.2314	−2.3990	−5.9997	−11.0258

against $\max_c |z_c|$ over ten uniformly random label permutations (`default_rng` seed 20260808) reading

$$0.9359,\ 1.6073,\ 1.9921,\ 1.5687,\ 2.1392,\ 1.1216,\ 1.7178,\ 1.1348,\ 3.2040,\ 1.3192,$$

mean 1.6741, never above 3.3. Ten is a sample and not an enumeration, and the true value ranking first among the eleven gives a Monte-Carlo p-value of 1/11, which is the smallest this many permutations can produce. It is not a bound on the exact permutation p-value, which ten draws cannot bound from above at all; the enumeration of Measurement 15 below is the stronger instrument, and this one's limit is the sample's. The correlation between depth and z_c is -0.9106 for the true labelling and $+0.0042$ on average under permutation. None of the ten sampled permutations reproduced either true value — which is a statement about the sample and not about the permutation law — so by the second half of Lemma 8 what is detected is the correspondence between cells and divisibility, not the cell sizes.

The same run corrects a natural reading of Lemma 2. One expects the floor se_c to be a property of the cell sizes, hence nearly invariant under permutation. It is not: the floor **collapses**, by factors from 3.8 at depth 5 to 105 at depth 0 — $1.2400 \cdot 10^{-1}$ against $1.1794 \cdot 10^{-3}$ there (`lab_mask_placebo.py`).

Note 11 (the floor is itself signal). The mechanism of that collapse is the three-term structure of (3). For a random subset of the band, $u_c(v) \approx (n_c/n)\, u_a(v)$, and then $Q_{cc}/n_c^2 - 2Q_{ca}/(n_c n) + Q_{aa}/n^2$ cancels to leading order, leaving only the fluctuation around proportionality. An arithmetic cell breaks that proportionality, and the floor is what is left over. **The exact floor is therefore not a noise level but a second measurement of the same correspondence.** It is not, however, a conservative version of some truer noise level. Lemma 2 is an identity: under the sign null the variance of the arithmetic cell's numerator *is* the exact floor, so $z_c = -11$ at depth 5 is exact under that null and not a discounted figure. The null itself is narrower than the arithmetic it stands in for: an i.i.d. sign ensemble on this object measured 0.6455 of the spread of a random *multiplicative* one — a model in the tradition of [8] — at 512 draws (`audit_cn_coin_deep.py`). The run prints that ratio to five places and quotes no sampling uncertainty for it; at 512 draws only the leading two figures are earned, and nothing below rests on more. That measurement rests on one band, one class family and one statistic, and its own registration says so: it is the second moment $E[Z^2]$ — the run writes G for Z — on the classes cut by $3, 5, 7$ over $(2^{20}, 2^{21}]$, not the first-order $m_c - \overline{m}$ on the depths cut by $3, 5, 7, 11, 13$ over the octave read here. No map between the two is established anywhere, so the direction is measured and the size is not: an ensemble that shares μ's multiplicativity is the wider of the two, and how much of the -11 that would take back is not fixed by anything in this paper. The conclusions below survive the direction; the word *calibrated* would not, and is not used. Dividing instead by the permuted floor (4), which at depth 5 is $1.1403 \cdot 10^{-1}$, gives $z \approx -42$. The neighbouring question — the numerator against the permutation null (5) — has the exact answer 42.42 (Note 14), and the two sit close because (4) is (5) averaged over the sign null. Reading them as one number quoted conservatively and boldly would be reading the count-based bar as the truth, and by Lemma 12 that bar is what (5) is: at depth 5 it is $1.1356 \cdot 10^{-1}$, which is $\mathrm{sd}(Z)/\sqrt{n_c}$ to five figures. The ten draws averaged $1.1518 \cdot 10^{-1}$, and that number is a sample where the two displayed are not. The sign null is the one this paper's statements are quoted against, throughout. The result files cited here name it `lem:coin`, after the statement of the generation they were written in; that name means this sign null and no further statement, and it is the null the width comparison earlier in this note is about.

Lemma 12 (the floor under the placebo, in closed form). *Let c be a cell of size n_c in a band of size n, let π be the uniformly random permutation of Lemma 8, and let c^π be c under the labelling $\ell \circ \pi$ — a uniformly random n_c-subset of the band. Write Var_ε for the variance under the sign null of Lemma 2 and E_π, Var_π for expectation and variance over π with the field Z held fixed. Then*

$$\tau_c^2 \;:=\; E_\pi \operatorname{Var}_\varepsilon\bigl(m_{c^\pi} - \overline{m}\bigr) \;=\; \frac{n - n_c}{n_c\,(n-1)}\Bigl(1 - \frac{Q_{aa}}{n^2}\Bigr), \tag{4}$$

and, for the field as observed,

$$\sigma_c^2 \;:=\; \operatorname{Var}_\pi\bigl(m_{c^\pi} - \overline{m}\bigr) \;=\; \frac{n - n_c}{n_c\,(n-1)}\, s^2, \qquad s^2 = \frac{1}{n}\sum_N \bigl(Z(N) - \overline{m}\bigr)^2, \tag{5}$$

with $E_\varepsilon s^2 = 1 - Q_{aa}/n^2$. Neither right-hand side sees the cell beyond its size: Q_{aa} is the one term of (3) that does not depend on c.

Proof. Put $g(N) = V(N)^{-1/2}$ and $K(N,N') = g(N)g(N')\sum_v \mu^2(v)\Lambda(N-v)\Lambda(N'-v)$, so that K is the covariance of Z under independent signs and $Q_{cd} = \sum_{N\in c}\sum_{N'\in d} K(N,N')$. The diagonal is

$$K(N,N) = \frac{\sum_v \mu^2(v)\Lambda(N-v)^2}{V(N)} = 1,$$

which is the normalisation of Z. It is worth saying why the equality survives the null. $V(N) = \sum_v \mu^2(v)\Lambda(N-v)^2$ is a function of the support of μ^2 and of Λ alone; the null replaces the signs $\mu(v)$ on that support and leaves the support itself fixed, so $V(N)$ is deterministic and unchanged, and the numerator above is literally $V(N)$. Every use of $K(N,N) = 1$ below — and the whole of this lemma rests on that diagonal — is therefore exact rather than approximate. With $J_c(N) = \mathbf{1}_c(N) - n_c/n$ one has $m_c - \overline{m} = n_c^{-1}\sum_N J_c(N)Z(N)$, so

$$\operatorname{Var}_\varepsilon\big(m_c - \overline{m}\big) = \frac{1}{n_c^2}\sum_{N,N'} J_c(N)J_c(N')K(N,N'), \tag{6}$$

which is (3) with its three terms collected. Under π the indicator of c^π is that of a uniform n_c-subset, so with $p = n_c/n$, and since $E_\pi J_{c^\pi} = 0$,

$$E_\pi\big[J_{c^\pi}(N)J_{c^\pi}(N')\big] = p(1-p)\Big(\delta_{NN'} - \frac{1-\delta_{NN'}}{n-1}\Big).$$

Taking E_π of (6) and using $\sum_N K(N,N) = n$ and $\sum_{N\neq N'} K(N,N') = Q_{aa} - n$,

$$E_\pi \operatorname{Var}_\varepsilon = \frac{p(1-p)}{n_c^2}\Big(n - \frac{Q_{aa}-n}{n-1}\Big) = \frac{p(1-p)}{n_c^2}\cdot\frac{n^2 - Q_{aa}}{n-1},$$

which is (4). For (5) repeat the computation with K replaced by the fixed matrix $(Z(N) - \overline{m})(Z(N') - \overline{m})$: its diagonal sums to ns^2 and its off-diagonal part to $-ns^2$, since $\sum_N (Z - \overline{m}) = 0$. Finally $E_\varepsilon \sum_N (Z - \overline{m})^2 = n - Q_{aa}/n$. The two are one identity: by Fubini, (4) is (5) averaged over the sign null — the without-replacement variance of a sample mean, with the population variance replaced by its sign-null value. □

Note 13 (what the collapse is). Lemma 12 makes the collapse of Measurement 10 a computation. For a random cell the floor is, in average over the permutation of the variance, the without-replacement formula and nothing else: it decays like $n_c^{-1/2}$, which is exactly the count-based reflex this paper began by rejecting, with $\operatorname{sd}(Z)$ replaced by $(1 - Q_{aa}/n^2)^{1/2}$ and a finite-population factor attached. That factor is $\sqrt{(n-n_c)/(n-1)}$ — the standard finite-population correction for a sample mean drawn without replacement [5] — and it is near 1 only for a cell small against its band: at this octave, where $n = 10^6$, the two shallow cells hold 383617 and 421978 of it and the factor reads 0.785 and 0.760, against 0.99997 at depth 5. At the octave $(2\cdot 10^6, 4\cdot 10^6]$, $n = 1000000$ and $1 - Q_{aa}/n^2 = 0.871282$ against $s^2 = 0.864029$, a ratio of 0.991676. **So the count-based scale is the Rademacher-null scale for a random cell and is not for the arithmetic cells cut here** — the thesis of this paper is a statement about which cells, not about counts. The two halves are not equally supported: the first is Lemma 12 and holds for every uniformly random subset, the second is Measurement 10 and covers the depth cells of one band. A cell cut some third way — by an interval of the band, or by the size of N — is neither, and nothing here says the bar holds there.

The exact floor of the arithmetic cell exceeds (4) by

depth	0	1	2	3	4	5
se_c/τ_c	104.80	42.68	70.28	44.37	16.97	3.83

— the factors quoted above, now read off a formula rather than off ten draws. Those ten draws stand to τ_c in the ratios 0.9968, 1.0195, 0.9991, 1.0175, 0.9992, 1.0101, and their variances to (4) itself in 0.9960, 1.0420, 0.9992, 1.0379, 0.9999, 1.0216; the largest departure is 1.12 of the draws' own standard error (`lab_mask_placebo.py`, L5). Ten draws and a formula agree, and it is the formula that is quoted.

Note 14 (the numerator's null is exact)*.* The second identity supplies what ten draws could not. Under $\ell \circ \pi$ the numerator $m_{c^\pi} - \overline{m}$ is the mean of a uniform n_c-subset of the fixed multiset $\{Z(N)\}$ minus its mean: mean 0, variance (5) exactly, no draw involved. At the same octave the true labelling's numerators sit at

depth	0	1	2	3	4	5
$m_c - \overline{m}$	+0.01325	+0.05191	−0.03432	−0.58002	−1.99506	−4.81664
$\lvert m_c - \overline{m}\rvert/\sigma_c$	11.25	47.71	16.33	106.89	102.22	42.42

standard deviations of the permutation null. Chebyshev's inequality, which needs nothing beyond (5), bounds the permutation p-value of the depth-5 numerator by $5.56 \cdot 10^{-4}$ and the union over the six cells by $1.28 \cdot 10^{-2}$. That replaces the $p \le 1/11$ of Measurement 10 by a computation *for the numerator*. It does not do so for the ratio z_c: the permutation law of the denominator is not given here, and Measurement 10's rank test remains the only statement about $\max_c |z_c|$ under permutation.

5 What predicts the floor's size

Write $\mathfrak{S}_2(h)$ for the Hardy–Littlewood singular series of the shift h — the local density of pairs $(p, p+h)$ — and, for a cell c in the band, put

$$E_{\mathrm{same},c}[\mathfrak{S}_2] = \frac{1}{n_c(n_c-1)} \sum_{\substack{N,N' \in c \\ N \ne N'}} \mathfrak{S}_2(N-N'), \qquad E_{\mathrm{all}}[\mathfrak{S}_2] = \frac{1}{n(n-1)} \sum_{\substack{N,N' \in a \\ N \ne N'}} \mathfrak{S}_2(N-N').$$

Both sums run over *ordered* pairs with the diagonal removed. The ordering is immaterial, since $\mathfrak{S}_2(-h) = \mathfrak{S}_2(h)$; the removal is not, since $\mathfrak{S}_2(0)$ is not defined, and every count and every denominator below is the off-diagonal one. Put

$$D_c := E_{\mathrm{same},c}[\mathfrak{S}_2] - E_{\mathrm{all}}[\mathfrak{S}_2]. \tag{7}$$

For the exact doubly centred analogue, also write

$$E_{c,a}[\mathfrak{S}_2] := \frac{1}{n_c(n-1)} \sum_{N \in c} \sum_{\substack{N' \in a \\ N' \ne N}} \mathfrak{S}_2(N-N'), \qquad B_c := E_{\mathrm{same},c}[\mathfrak{S}_2] - 2E_{c,a}[\mathfrak{S}_2] + E_{\mathrm{all}}[\mathfrak{S}_2].$$

Thus B_c, not D_c, is the off-diagonal singular-series quantity with the same three-part centring as (6). The two-term D_c is retained as the descriptor used by the measurements below; Measurement 15 quantifies how close it is to the doubly centred convention in the measured band.

Measurement 15 (the floor tracks D_c, including its shape)**.** At the octave $(2 \cdot 10^6, 4 \cdot 10^6]$:

depth	0	1	2	3	4	5
D_c	0.302138	0.046766	0.412504	1.052129	1.946109	3.171298
se_c	0.12400	0.04662	0.14834	0.24178	0.33253	0.43685

with correlation 0.9805 across depths. Neither row is monotone: both dip at depth 1, and that is the mechanism made visible. Depth 0 is the cell on which *none* of $3, 5, 7, 11, 13$ divides N, and excluding a class concentrates the shift just as fixing one does — if $3 \nmid N$ and $3 \nmid N'$ then both lie in $\{1, 2\}$ mod 3 and $3 \mid h$ with probability $\frac{1}{2}$ against $\frac{1}{3}$ for a random pair. Both ends of the depth index concentrate h; depth 1 mixes the patterns and washes out. Depth is therefore a coarse index for D_c, and the floor follows the same coarse index (`lab_cell_singular.py`). The control is that run's own, and is an enumeration rather than a sample: over all 720 permutations of the depth labels exactly one reaches $|r| = 0.9805$, against a median of 0.3475 and a 95th percentile of 0.8084 over that set. Those are order statistics of the 720 relabellings and not quantiles of a law: the depths are deterministic, no exchangeability of the labels is assumed, and what the enumeration measures is how much of the agreement is carried by which cell gets which number rather than by the shape of either row alone. The label permutation of Lemma 8 does not supply it, since that one permutes the labels of z_c and not of this pair.

A correlation over six points cannot tell se_c from any other quantity that falls with the depth, and the cell count is one of those. The ratio settles it and needs no null: se_c^2/D_c reads 0.05089, 0.04648, 0.05334, 0.05556, 0.05682 and 0.06018 — a full width of 25.4% of its mean, which is the figure the result file prints, and from depth 1 upward a monotone rise of 29% over its smallest value — while D_c itself moves by a factor 68 across the same six cells. That rise is not a property of the ratio. se_c^2 carries the diagonal of K, where $K(N, N) = 1$, and D_c does not: its averages run over pairs $N \neq N'$, and $\mathfrak{S}_2(0)$ is not defined. The diagonal is $n_c^{-2} \sum_N J_c(N)^2 = (1 - n_c/n)/n_c$ and has no partner in the denominator. At the deepest cell it is 7.82% of se_c^2, and what it contributes there, +0.004706, exceeds the whole printed rise from depth 4 to depth 5, +0.00336. That contribution is depth 5's own share and not the step between the two depths; the cited run states it as such.

No single factor carries se_c^2 to D_c's convention. (6) is the doubly centred mean $E_{\mathrm{same},c}[K] - 2E_{c,a}[K] + E_{\mathrm{all}}[K]$, which is three sums over three index sets with three different pair counts — n_c^2, $n_c n$, n^2. Each term has to lose its own diagonal and be divided by its own count, and $K(N, N) = 1$ makes that exact. Done that way the row reads 0.05089, 0.04646, 0.05334, 0.05553, 0.05665, 0.05704: a full width of 19.8% rather than 25.4%, monotone from depth 1 still, and a step from depth 4 to depth 5 of +0.00038. **The rise survives the match; what shrinks is its size** — nine times smaller at the last step, and a rise of 23% from depth 1 rather than 29%. Removing only the diagonal gives −0.00112 there and scaling that by $n_c/(n_c - 1)$ gives −0.00031; both are over-corrections and neither is the comparison to make. The exact doubly centred comparison is between two kernels: (6) is the doubly centred mean $n_c^{-2} \sum J_c J_c K$, while B_c is the corresponding off-diagonal doubly centred mean with K replaced by the singular series of the shift. The descriptor D_c in (7) omits the cross-centering term and is therefore not literally the same functional. The centring is not cosmetic. For K the cross term is not the band mean — it runs 0.124636 to 0.140049 against $E_{\mathrm{all}} = 0.128718$, and dropping it would misstate the variance by factors 0.469 to 2.120 across these six depths (`lab_mask_placebo.py`). For the singular series it is: $E_{c,a}$ departs from E_{all} by at most 0.004192, so the two-term difference (7) and the doubly centred form agree to within one per cent at five of the six depths — the largest of those five being 0.59% at depth 3 — and to within five per cent at depth 1, where D_c is smallest (`lab_cell_singular.py`). Both bands quoted here have the same numerator, the three-term variance se_c^2, and differ only in the denominator: against D_c as (7) defines it the band is the 25.4% above, and against the doubly centred B_c it is 29.7%. A near-constant ratio through the origin is what would follow if K and the singular series differed across the band by a factor that the double centring leaves alone. That is not proved here: the ratio is measured, not derived. What the measurement does settle is the alternative it was taken against — a quantity that merely decreases with the depth produces no such ratio. What it does not settle is the other direction. The band is read across depths at one octave; across scale the same ratio falls over the three octaves on which D_c is measured. Divided by the sampled D_c it reads 7.9%, 0.9%, 9.4%, 9.2%, 9.6%, 15.2%, and the 0.9% at depth 1 was read here as that depth being the exception. It is not: divided by the D_c this measurement computes over all pairs, the six read

9.5%, 9.6%, 9.5%, 9.6%, 9.8%, 15.3%. **The exception was the estimator's** — at depth 1 the pair average carries more noise per octave than the drift it was said to show. What remains is a drift of about half the matched width at five of the six cells and rather more at the deepest. Both inputs are printed here (`lab_cell_singular.py`, which forms the quotient from its own D_c and the variance row of `lab_cell_floor.txt`). So D_c fixes the ratio's depth profile at a fixed scale and not its scale dependence, and the floor itself is not scale-invariant (Measurement 6 fits it a nonzero exponent), while Proposition 17 shows that D_c is asymptotically scale-free: D_c is therefore not all of what fixes the floor's size.

The proof of the proposition below sums a weight along an arithmetic progression. What it needs of the weight is only bounded variation, and that is worth isolating, because it is the one step where the shape of the weight — as opposed to its arithmetic — does any work.

Lemma 16 (summing along a progression against an envelope)**.** *Let $w : \mathbb{Z} \to \mathbb{R}$ be supported on $|h| < B$ with total variation $\mathrm{TV}(w) = \sum_h |w(h+1) - w(h)|$, and let $q \geq 1$, $a \in \mathbb{Z}$. Then*

$$\sum_{h \equiv a\ (q)} w(h) \;=\; \frac{1}{q} \sum_h w(h) \;+\; O\big(\mathrm{TV}(w)\big),$$

with an absolute implied constant — in particular uniform in q, in a, and in B.

Proof. Partition all of $\mathbb{Z}$ — not the support — into the blocks I_j of q consecutive integers determined by a, and let h_j be the representative of the class in I_j. Every block is then complete, and all but finitely many meet the support not at all and contribute zero to both sides. So

$$\sum_{h \equiv a\,(q)} w(h) - \frac{1}{q}\sum_h w(h) \;=\; \sum_j \Big[w(h_j) - \frac{1}{q}\sum_{h\in I_j} w(h) \Big] \;=\; \sum_j \frac{1}{q} \sum_{h \in I_j} \big[w(h_j) - w(h) \big],$$

and $|w(h_j) - w(h)| \leq V_j$ for every $h \in I_j$, where V_j is the variation of w within I_j. The inner average is therefore at most V_j, and the V_j are sums of $|w(h+1) - w(h)|$ over disjoint sets of h, so $\sum_j V_j \leq \mathrm{TV}(w)$. The implied constant is 1. □

Nothing in it is arithmetic: q is arbitrary, and the bound does not improve with q nor degrade with it. That is exactly why it survives the primes comparable to the band length, which are equidistributed modulo nothing.

Proposition 17 (the size mechanism, asymptotically scale-free)**.** *We establish an asymptotic statement and not an identity. Let $Q = 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13$, let a be the even integers of an interval of length B, and put $\ell = B/2Q$, so that each of the Q even residue classes modulo $2Q$ meets a in $\ell + O(1)$ terms and each odd class meets it in none. Let R_c be a nonempty set of m even residue classes modulo $2Q$ and $c = \{N \in a :\ N \bmod 2Q \in R_c\}$ the cell they cut. Every depth cell is of this form — depth fixes how many of the five primes divide N and not the residues themselves, so R_c is the set of classes of the $\binom{5}{d}$ divisibility patterns of depth d — and a itself is the case in which R_c is all Q even classes. Write*

$$A(h) = \prod_{2<p|h,\ p\leq 13} \Big(1 + \frac{1}{p-2}\Big), \qquad C_2 = \prod_{p>2} \frac{p(p-2)}{(p-1)^2}, \qquad \kappa = 2C_2 \prod_{p>13} \Big(1 + \frac{1}{p(p-2)}\Big),$$

so that A is a function of $h \bmod Q$, and put

$$\mathcal{A}_c = \frac{1}{m^2} \sum_{r,r' \in R_c} A(r - r'), \qquad \mathcal{A}_a = \frac{1}{Q^2} \sum_{r,r'} A(r - r'),$$

the second sum over all pairs of even classes modulo $2Q$. These are rational numbers determined by R_c alone: they depend neither on B nor on where the interval sits. Then, with D_c as in (7) and for every fixed $\eta \in (0,1)$,

$$D_c \;=\; \kappa\,(\mathcal{A}_c - \mathcal{A}_a) \;+\; O\Big(\frac{1}{\ell}\Big) \;+\; O_\eta\Big(\frac{B^\eta}{\ell}\Big) \;+\; O_\eta\big(B^{\eta-1}\big). \tag{8}$$

The implied constant in the first error term is absolute and those in the other two depend on η alone; in particular none of them depends on B or on the position of the interval, and the parameter set $\{3,5,7,11,13\}$ enters only through Q, fixed throughout. The first constant is absolute in the sense of being bounded over the cells and not of being one number for all of them: the rounding residue isolated in the proof carries a constant that varies by two orders of magnitude between depth 0 and depth 4, and it is the finiteness of the field — Q fixed, hence finitely many cells — that makes the supremum a constant at all. Nothing below needs the stronger reading. The η-dependence is not uniform as $\eta \to 0$ — the constant comes from the Euler products of Rankin's bound in the proof and blows up there — so η is fixed in advance and the statement is one about each fixed η separately.

Consequently $D_c \to \kappa(\mathcal{A}_c - \mathcal{A}_a)$ as $B \to \infty$ with R_c held fixed. Writing $D_c(B)$ for the value of (7) in a band of length B, define the one-step exponent

$$e_c(B) \;=\; -\,\frac{\log\big|D_c(2B)/D_c(B)\big|}{\log 2}. \tag{9}$$

Where $\mathcal{A}_c \neq \mathcal{A}_a$ the convergence gives $e_c(B) \to 0$; where $\mathcal{A}_c = \mathcal{A}_a$ it gives only $D_c \to 0$ and predicts no exponent. No rate is claimed for either limit, and nothing here asserts equality at a fixed band. Measurement 18 writes $\mathcal{A}_c$ and $\mathcal{A}_a$ as $E_{\mathrm{same},c}[A]$ and $E_{\mathrm{all}}[A]$, and Measurement 19 writes the first $E_c^{\mathrm{ex}}[A_S]$; the pair average of A over the cell itself differs from $\mathcal{A}_c$ by the first error term.

Proof. For N, N' in a band, $\mathfrak{S}_2(N-N')$ depends on $N-N'$ only through the local factors at primes dividing $h = N - N'$. Depth does *not* fix the residue of N modulo each of $3,5,7,11,13$ — it fixes only how many of them divide N, so a depth-d cell is the union of the $\binom{5}{d}$ divisibility patterns of that size, and Measurement 15 records the mixing this causes. What the argument needs is weaker and does hold: each pattern fixes, at each of the five primes, the admissible residue set, so within a pattern the law of h modulo each of them is determined; and, the even N of a band being equidistributed modulo $3\cdot5\cdot7\cdot11\cdot13$ to $O(1/\ell)$ per class, the joint residue-class structure within a pattern is determined by the product modulus. The word is meant in the Chinese-remainder sense and not the probabilistic one: the admissible set modulo $3\cdot5\cdot7\cdot11\cdot13$ is the product of the admissible sets at the five primes, so the joint law modulo the product is determined and not only the five marginals. No independence of random variables is asserted or used. The weights with which the patterns enter a depth cell are their densities among the even N of the band, which are the same at every scale. The induced law of h modulo those primes is therefore determined by the cell alone up to the resolution of the cell itself. That resolution is $O(1/\ell)$ and not $O(1/n_c)$: a depth cell meets each joint residue class in about $\ell = B/2Q$ terms, and it is that count, not the cell's total, that the empirical law of the residues is read from. The same counting governs the divisor sampling below — the resolution is $O(1/\ell)$ for both and not $O(1/n_c)$ — and no cancellation is claimed for either.

Write $Q = 3\cdot5\cdot7\cdot11\cdot13$ and split $\mathfrak{S}_2(h) = 2C_2\,A(h)\,T(h)$, where $A(h) = \prod_{2<p|h,\ p\le 13}(1+\frac{1}{p-2})$ is a function of $h \bmod Q$ and $T(h) = \prod_{p|h,\ p>13}(1+\frac{1}{p-2})$ carries everything above 13. What has to be shown is that T contributes the same factor to both averages of (7), whatever the cell. That is the hardest step of the proof and not a hypothesis readable off the construction. It is not enough that each prime above 13 is separately equidistributed: $\mathfrak{S}_2$ reads them all at once, and the primes comparable to the band length are equidistributed modulo nothing.

Expand T over divisors. Writing $g(d) = \prod_{p|d} \frac{1}{p-2}$ and letting d run over the squarefree integers all of whose prime factors exceed 13, $T(h) = \sum_{d|h} g(d)$. Every such d is coprime to $2Q$; and the shifts that occur are $h = N - N'$ with both N even, so they lie in classes modulo $2Q$ and not modulo Q. For a residue a modulo $2Q$ the Chinese remainder theorem gives

$$\#\{h : 0 < |h| < B,\ h \equiv a\ (2Q),\ d \mid h\} = \frac{B}{Qd} + O(1),$$

and $d > B$ divides no h of the band at all, since $h = N - N'$ with both N inside a band of length B.

The weight with which h enters a pair average is not a function of $|h|$ alone. For $E_{\mathrm{same},c}$ it counts the $N \in c$ with $N - h \in c$, and membership of c is fixed by $N \bmod 2Q$, so the count reads $h \bmod 2Q$ as well: if $3 \mid h$ the shift preserves divisibility by 3 and otherwise it does not. It does not, however, factor. Write $f_c(h) = \#\big(R_c \cap (R_c + h)\big)$ for the number of classes that a shift by h keeps inside R_c. The N of one such class that are admissible fill an interval of length $B - |h|$, which the class meets $(B - |h|)/2Q$ times up to the rounding at its two ends, so

$$W_c(h) \ =\ \#\{N \in c:\ N - h \in c\} \ =\ f_c(h)\,\frac{B - |h|}{2Q} \ +\ \rho_c(h), \qquad |\rho_c(h)| \ \le\ f_c(h) \ \le\ m.$$

The residue ρ_c is that rounding summed over the surviving classes. It is additive rather than a further factor, and it is bounded independently of B.

The unweighted ratio is $O(1/\ell)$, but the pair average carries $\rho_c(h)$ against $T(h)$, and T is not uniformly bounded. The weighted residue is instead absorbed by the already displayed $O_\eta(B^{\eta-1})$ term. Indeed

$$T(h) = \sum_{d|h} g(d) \le \tau(|h|), \qquad \sum_{0<|h|<B} |\rho_c(h)| T(h) \le m \sum_{0<|h|<B} \tau(|h|) = O_\eta(mB^{1+\eta}).$$

After division by the $m^2\ell^2$ ordered pairs of a cell, this is $O_\eta(B^{\eta-1})$ uniformly in the finitely many cells (with Q fixed). Thus no unweighted estimate is used after multiplication by T.

What the argument uses is therefore the leading product, with the weighted ρ_c contribution carried into $O_\eta(B^{\eta-1})$. Its second factor is an envelope in $|h|$: normalised to sum 1 it is $(B - |h|)/Z$ with $Z = \sum_h (B - |h|)$, so consecutive values differ by $2/Z$ and there are $O(B)$ of those steps, whence total variation $O(1/B)$ — with constant 4 when h runs over the even integers, since then $Z = B^2/2 + O(B)$. Lemma 16 applied to it gives $P(d \mid h) = 1/d + O(1/B)$ uniformly in $d \le B$. The mixture over classes is what the pattern weights supply. Every cell's pair law is such a mixture, whence

$$E[\mathfrak{S}_2] = 2C_2\, E[A] \prod_{p>13} \Big(1 + \frac{1}{p(p-2)}\Big) \ +\ \mathcal{E}, \qquad \mathcal{E} = O(1/\ell) + O_\eta\big(B^\eta/\ell\big) + O_\eta\big(B^{\eta-1}\big) \ \ (\eta > 0).$$

The three parts of $\mathcal{E}$ are derived in the paragraphs that follow: the first from counting a cell against its residue classes, the second by summing that error against g, and the third from the weighted rounding residue and the two truncations of the divisor sum. They are stated here and justified there rather than the other way round, so that the shape of the answer is visible before the bookkeeping.

All three parts of $\mathcal{E}$ have to be said. None of their *rates* depends on the cell: $\ell = B/2Q$ is a band parameter, the same at depth 0 and at depth 5. The constants in front of them are the separate question answered above for the first — bounded over the cells because Q is fixed, and not one number for all of them. A cell-dependent bound would have to be an $O(1/n_c)$, and the counting below does not support one. At the level of the displayed powers, the divisor-sampling bound is B^η times the first, while $B^\eta/\ell = 2Q\,B^{\eta-1}$, so the latter two bounds have the same asymptotic order for fixed Q. Their finite-band numerical sizes cannot be ordered without the unnamed implied constants, and the conclusion does not require such an ordering. The $O(1/B)$ above is the error of the envelope, not of the cell. A cell is not a random sample: it is a union of residue classes modulo $2Q$, and every d here is coprime to $2Q$, so the count that served the envelope serves the cell as well — taken with the cell's own parameters. A depth cell is m classes, each meeting the band in $\ell = B/2Q$ terms, with $m\ell = n_c$; only the deepest cell of this field is a single class. Over one pair of classes the pairs with $d \mid h$ number $\ell^2/d + O(\ell)$ — the count is a tent summed along a progression, and its uniformity in d is the paragraph on uniformity below — so over the m^2 pairs of classes the error is $O(m^2\ell) = O(n_c^2/\ell)$ and the proportion departs from $1/d$ by $O(1/\ell) = O(Q/B)$ — uniformly in d, with no gain from the size of d. Summed against g that is $\ell^{-1}\sum_{d\le B} g(d)$, which the Rankin bound below puts at $O_\eta(B^\eta/\ell)$; ℓ is a fixed fraction of B — namely $B/2Q$ with Q fixed, so that $B^\eta/\ell = 2Q\,B^{\eta-1}$ as recorded above — so this is $O_\eta(B^{\eta-1})$ again.

The counting gives $O(1/\ell)$ and not $O(1/n_c)$, and the two differ by the factor m, which is large at every cell but the deepest. The stronger form would need the m^2 class-wise errors to cancel rather than accumulate. Nothing here establishes that, and it is not claimed. This is the same $O(1/\ell)$ that the equidistribution above carries, arrived at by the same count of a cell against its residue classes; the two are one resolution seen twice and not two errors. Both are bounds rather than errors, and their finite-band numerical ordering is not supplied because the implied constants are unnamed. Neither varies by cell. The conclusion needs no ordering, only that both go to zero, and ℓ grows with the band.

The last part is the two truncations, which Rankin's trick bounds. For $s > 0$, $\sum_{d\le B} g(d) \le B^s\prod_{p>13}(1+\frac{1}{(p-2)p^s})$, the product converging because $\sum_p p^{-1-s}$ does; this is $O_\eta(B^\eta)$ at $s=\eta$, and reaches $O_\eta(B^{\eta-1})$ only after the per-d error $O(1/B)$ multiplies it. For $t<1$, $\sum_{d>B} g(d)/d \le B^{-t}\prod_{p>13}(1+\frac{1}{(p-2)p^{1-t}})$, the product converging because $\sum_p p^{-2+t}$ does; at $t = 1-\eta$ this is $O_\eta(B^{\eta-1})$ directly. The two reach the same order by different routes, and the two exponents cannot be collapsed into one: with $s=t$ the bounds read B^{s-1} and B^{-s}, which meet only at $s=\frac{1}{2}$, and $B^{-1/2}$ is far weaker than $B^{\eta-1}$. Nor does η pay to be taken small without limit: log of the product is of order $\log(1/\eta)$, so the bound is best near $\eta = 1/\log B$ and worsens on either side of it. The two exponents stay two, but at $s=\eta$ and $t=1-\eta$ the two *products* coincide: both read $\prod_{p>13}(1+\frac{1}{(p-2)p^\eta})$, and the two convergence criteria both reduce to $\sum_p p^{-1-\eta}$. There is one η-constant in this proof, not two.

Every constant here is uniform in the cell, which is what lets $\mathcal{E}$ be one bound for a whole band. A is exactly constant on each pair of residue classes modulo $2Q$ — h is fixed there — and is bounded by $\prod_{2<p\le 13}(1+\frac{1}{p-2}) = 128/33$, a number with no cell in it. On one class pair the ordered pairs (i,j) in $[0,\ell)^2$ with $i-j$ in a fixed class modulo d number $\ell^2/d + O(\ell)$ with an absolute constant, since the difference count is a tent of total variation 2ℓ. And the m^2 class pairs divide out identically against $n_c^2 = m^2\ell^2$: the same relative error appears at $m = 5760$ and at $m=1$. Nothing in the bound names the cell's pattern set.

What is *not* uniform is $\mathcal{E}/D_c$, and it is that ratio the exponent needs. At the octave Measurement 20 reads, D_c runs from 0.046766 at depth 1 to 3.171298 at depth 5, a factor of 67.8; one and the same $\mathcal{E}$ therefore buys 67.8 times less at the first than at the last, at every band. So the proposition is not equally informative at the six cells, and depth 1 is the cell where it says least — the same depth whose main term the proof cannot certify is nonzero.

The factor the primes above 13 contribute is therefore one and the same constant $\kappa/2C_2$ in both averages, and it leaves the difference of the two pair averages of A. Each of those is the average of a function of $h \bmod Q$ over the pairs of a union of residue classes, and the classes meet the band in $\ell + O(1)$ terms each, so each pair average is $\mathcal{A}_c$, respectively $\mathcal{A}_a$, up to $O(1/\ell)$ with an absolute constant, A being bounded by 128/33. Hence

$$D_c = \kappa\,(\mathcal{A}_c - \mathcal{A}_a) + \mathcal{E},$$

which is (8). The main term is fixed by R_c and is the same at every scale, so (7) converges to it. Convergence is additive and an exponent is multiplicative: if the limit is nonzero then $D_c(2B)/D_c(B) \to 1$ and the one-step exponent (9) tends to 0, while if $\mathcal{A}_c = \mathcal{A}_a$ the proposition says only that $D_c \to 0$ and predicts no exponent at all. No derivative is taken anywhere: D_c is defined band by band, and what the convergence controls is the ratio of consecutive bands, which is what (9) reads. The difference $\mathcal{A}_c - \mathcal{A}_a$ is a finite sum of rationals over the residues modulo $2Q$ and the pattern weights, so it is decidable cell by cell without any measurement. Measurement 18 decides it: the difference is a nonzero rational at all six depths, so the exponent this proposition predicts is defined at all six. $\square$

Measurement 18 (the main term, computed)**.** $A(h)$ depends only on which of $3, 5, 7, 11, 13$ divide h, a cell is a union of residue classes modulo $2Q$, and $p \mid h$ for h between two classes exactly when the two agree modulo p. So $E_{\mathrm{same},c}[A] - E_{\mathrm{all}}[A]$ is a finite double sum over pattern pairs weighted by $m(S) = \prod_{p \notin S}(p-1)$, and it is computed in exact rational arithmetic (`lab_maincoef.py`): 635653709/2854051200, 10743253/313945632, 318004704/1037775739, 296100736/378194817, 13823072/9546537 and 78976/33033 at depths 0 to 5. **None is zero**, which is the hypothesis the proof needs. At depth 5 the cell is a single class, so Q divides every shift in it and $E_{\mathrm{same},c}[A]$ is exactly $\prod_{2<p\leq 13}(1+\frac{1}{p-2}) = 128/33$.

Multiplied by $2C_2 \prod_{p>13}(1+\frac{1}{p(p-2)})$ — a constant with no fitted parameter in it, and one that is not irrational: the same term-by-term cancellation that gives $E_\infty[\mathfrak{S}_2] = 2$ below reduces it to

$$2C_2 \prod_{p>13}\left(1+\frac{1}{p(p-2)}\right) = 2 \prod_{p\in\{3,5,7,11,13\}} \frac{p(p-2)}{(p-1)^2} = \frac{11011}{8192} = 1.3441162109375$$

exactly — these predict D_c at the octave $(2\cdot 10^6, 4\cdot 10^6]$:

depth	0	1	2	3	4	5
predicted	0.299361	0.045996	0.411876	1.052351	1.946236	3.213542
measured	0.302138	0.046766	0.412504	1.052129	1.946109	3.171298
residual/se	2.0	0.6	0.5	0.2	0.1	260.8

Six numbers are predicted from none, across cells spanning a factor of 68. *Four of the six sit inside two sampling errors of the measured value*, depth 0 sits at 2.0, and the sixth is depth 5 at 260.8. Both of the last two are accounted for below and neither by a fitted quantity. Over all 720 relabellings of the depth labels the count of cells agreeing within two sampling errors reaches the observed four in two of them — a fraction 2/720 of the enumerated set, read as in Measurement 15's control and not as a p-value under a law that produced the depths; the six predictions span a factor of 70, so a wrong pairing is refused by size alone, and what the enumeration shows is that the agreement is carried by which cell gets which number.

Depth 5's miss is not a residual but a computable quantity, and computing it is what the cell's structure allows. That cell is a single residue class modulo $2Q$, hence an arithmetic progression segment $N_0 + 2Qi$; every same-cell shift is $h = 2Qk$ with $0 < |k| < n_c$; and every d in the divisor expansion is coprime to $2Q$, so $d \mid h$ exactly when $d \mid k$. *No $d \geq n_c$ divides any same-cell shift at all*, while the Euler factor sums over every d. The resulting deficit is a finite sum, $\Delta = 8.505279 \cdot 10^{-3}$ at $n_c = 67$, and $2C_2 \cdot \frac{128}{33} \cdot \Delta = 0.043558$ against an observed residual of 0.042244 — agreement to 3.1 per cent, from a computation with no free parameter and no asymptotics. It takes depth 5 from 260.8 sampling errors to 8.1.

Calling that miss "the size of $1/\ell$" would instead be the reading the discussion below withdraws, and unfalsifiable besides: with an unnamed implied constant no discrepancy could refuse it.

It is also the same quantity Measurement 20 records as depth 5's spread. Δ falls with n_c — it is $1.297902 \cdot 10^{-2}$, $8.505279 \cdot 10^{-3}$ and $5.208268 \cdot 10^{-3}$ at $n_c = 34, 67, 134$ — so it induces a drift in D_5 across the three octaves of 1.24 per cent, against the 1.28 that measurement finds. *The two are not on one denominator*: the first is taken over the uncorrected prediction, the second over a mean of measured values, and the cited run prints both readings rather than one. What the comparison supports is the order of the effect, not an agreement to the second digit. *So the largest of the six spreads is not a failure of scale invariance: it is the proposition's own error term, computed.* An independent reading of the construction reaches both figures without this computation, putting the residual at 1.35 per cent and claiming only the order for the spread. The packet does not assert which was carried out first.

The constant cuts the singular series at 13, and an off-by-one in which primes are absorbed into C_2 would move it by 1/143, or 0.70 per cent. It is not there: such an error is a uniform multiplicative factor, and one of that size would show at 11.5 sampling errors at depth 4 alone.

The band average has a limiting value in closed form, and the run estimated it instead. Write $E_{\mathrm{all},B}$ for the pair average over the band of length B — the quantity (7) subtracts — and E_∞ for its limit. The average of the singular series over shifts is classical [3], and the error term in it has been sharpened since [4]; what is needed here is only the leading constant. That constant is 1 when the average is taken over *all* shifts, and the average needed here is twice it. The reason is that $\mathfrak{S}_2(h) = 0$ for odd h — the local factor at 2 vanishes, since n and $n + h$ cannot both be odd primes above 2 — while every shift occurring here is even, $h = N - N'$ with N and N' both even. Averaging a function that vanishes on half the integers over the half where it does not doubles its mean. Taking $P(p \mid h) = 1/p$ for every odd p gives $E_\infty[\mathfrak{S}_2] = 2C_2 \prod_{p>2}(p-1)^2/(p(p-2))$, and $C_2 = \prod_{p>2} p(p-2)/(p-1)^2$ is the reciprocal of that product term by term, so

$$E_\infty[\mathfrak{S}_2] = 2.$$

This is a limiting value and not the band's own average, which is 1.999986 at the band Measurement 19 reads; the two differ by $1.4 \cdot 10^{-5}$, and that difference is the band's equidistribution error, discussed there.

The run's $E_{\mathrm{all},B}$, recoverable from its own table as $E_{\mathrm{same},c} - D_c$ and the same number in all six rows, is 1.998755: low against the limit by $1.2 \cdot 10^{-3}$, which is a hundred times the finite-band difference and is therefore the *sampling* error of that average. It is a single additive offset shared by every D_c, since each is $E_{\mathrm{same},c}$ minus that one number; it is one sampling error at five cells and eight at the sixth, because depth 5's 4422 pairs are enumerable where 10^{12} are not. It is not a truncated prime product: the sieve here runs over every prime to $8 \cdot 10^6$, and a truncation would in any case leave depth 5 exact, every same-cell shift there being free of prime factors above 61.

Substituting the limiting value for the sampled one — writing $D'_c := E_{\mathrm{same},c}[\mathfrak{S}_2] - 2$, which is (7) only up to the band's own $1.4 \cdot 10^{-5}$ — the prediction reads $\frac{11011}{8192} E_{\mathrm{same},c}[A] - 2$, one estimated input fewer, and the row closes:

depth	0	1	2	3	4	5
$D'_c = E_{\mathrm{same},c} - 2$	0.300893	0.045521	0.411259	1.050884	1.944864	3.170053
predicted	0.299361	0.045996	0.411876	1.052351	1.946236	3.169984
residual/se	−1.13	+0.36	+0.45	+1.07	+1.16	−0.43

All six sit inside 1.2 sampling errors, and nothing is fitted: the 2 is a theorem, the depth-5 entry carries the finite divisor deficit above, and the constant is the singular series' own. Depth 5 has gone from 260.8 to −0.43.

The computation is post hoc as to timing: it was written after the D_c row was measured. It is not a fit — the difference is forced by the residue structure and the constant by the definition of the singular series — but a registered rule of it was refuted. The rule predicted the worst relative deviation at depth 5; it is at depth 1 (1.65 per cent). **The rule was written on the wrong statistic**: the measured D_c is a sampled estimator whose sampling error this repository already prints beside it, and against that error depth 1 is 0.6 and depth 5 is 260.8. Read on the right quantity the claim holds; read as registered it does not, and it is recorded refuted.

Measurement 19 (a second cell shape, computed from residues alone)**.** Everything above is a statement about one partition. The computation of Measurement 18 runs for any set of odd primes, so a second partition can be *computed without reference to its own measurement*, which the first can no longer be given.

Cells indexed by depth over $S = (3, 5, 7)$, so $Q' = 105$ and $\ell = B/2Q'$ is 9524 at this octave against the first partition's 66.6. The constant is $K = 2\prod_{p\in S} p(p-2)/(p-1)^2 = 175/128$, exactly rational, and the predictions $K\,E_{\mathrm{same},c}[A_S] - 2$ are computed from the residues alone (`lab_secondcell_predict.py`), taking no input from the measured partition. The measurement (`lab_secondcell2.py`, registered in `lab_secondcell2_predict.py`) is exhaustive: a cell is a set of positions in the band, so the pair average of $\mathfrak{S}_2$ over it is one autocorrelation of the cell indicator, with no sampling, no seed and no standard error. The pair counts reproduce $n_c(n_c-1)$ exactly at all four cells.

depth	0	1	2	3
m	48	44	12	1
main term, uncorrected	0.295736	0.126485	0.825521	2.375000
fully corrected	0.295286	0.126068	0.824967	2.374142
predicted, $1/m$ corrected	0.295726	0.126476	0.825475	2.374142
exact	0.295708	0.126454	0.825436	2.374142
difference	-0.000018	-0.000022	-0.000039	-10^{-7}

The first two rows are the endpoints T1 gates on and are printed so that the rule can be checked from the table rather than from the file: the deficit is negative, so the interval at each depth is [fully corrected, uncorrected]. The row the third line predicts is neither endpoint — it is the $1/m$ diluted correction, which is what T2 scores. Of the three registered rules two hold and one is void. The exact value lies between the main term and the fully corrected prediction at all three of the cells where the two differ, which is the rule that gates, and the fitted exponent stays inside the spread the third rule allowed. **So the computation transfers to a cell shape it was not built on.** Depth 3 is a consistency check and not a score: there the prediction and the measurement are the same finite sum, as the registration records.

One rule was registered against the evidence then available. The sampled thirty-draw mean at depth 2 was 0.825796, above T1's upper bound of 0.825521 by about 1.2 standard errors of that mean, and the registration file records both that lean and that it was not decisive. The exhaustive finite-sum value is inside the interval. So that is one pass against a stated lean — not two, and not against a prediction of failure: the registration file says "leans, not decisive" of the first and "may fail" of the second, and the second is void rather than passed.

The correction the prediction carries should not be modelled at all, and this measurement is what shows it. The prediction diluted the divisor deficit by m, the number of residue classes a cell unions. Fitting an exponent instead, (undiluted)$/m^{\alpha}$, gives 0.715, 0.689 and 0.756 at $m = 48, 44, 12$. The registration fixed in advance that a value strictly between 0 and 1 says the model is wrong rather than that the exponent is that value, so **no exponent is claimed**: a dilution exists and $1/m$ is not it.

It does not need to. The deficit is the same autocorrelation with $T = \prod_{p|h,\,p>7}(p-1)/(p-2)$ in place of $\mathfrak{S}_2$, so it is computed rather than fitted — and computing it exposes a fourth term that no fit could have separated:

$$D'_c = \underbrace{K\,E^{\mathrm{ex}}_c[A_S] - 2}_{\text{main term}} + \underbrace{K\big(E_c[A_S] - E^{\mathrm{ex}}_c[A_S]\big)}_{\text{pattern weight}} - \underbrace{2C_2E_c[A_S]\Delta_c}_{\text{divisor deficit}} + \underbrace{2C_2\mathrm{Cov}_c(A_S, T)}_{\text{covariance}},$$

where E_c^{ex} is the exact rational the prediction is built from and E_c the measured pair average.

depth	0	1	2	3
main term	0.2957357	0.1264850	0.8255208	2.3750000
pattern weight	$-4.55 \cdot 10^{-6}$	$-6.75 \cdot 10^{-6}$	$-3.99 \cdot 10^{-6}$	0
deficit	$-1.32 \cdot 10^{-5}$	$-1.23 \cdot 10^{-5}$	$-5.35 \cdot 10^{-5}$	$-8.58 \cdot 10^{-4}$
covariance	$-1.02 \cdot 10^{-5}$	$-1.17 \cdot 10^{-5}$	$-2.69 \cdot 10^{-5}$	0
sum	0.2957078	0.1264543	0.8254364	2.3741421

and the sum reproduces the measured value at every cell. No exponent and no free parameter survive anywhere in it.

The pattern-weight term is the departure of the empirical class weights from the idealised ones — which is Proposition 17's *other* error term, the equidistribution one, measured here for the first time — and the covariance is the failure of $E[A_S T]$ to factor; **both are exactly zero at a single-class cell, which is the only kind either partition ever calibrated the deficit on**, so a fitted exponent was carrying two quantities that cannot appear where it was calibrated.

No power law in m fits these three, which is a firmer reason to drop the exponent than the absence of a mechanism. The exponent implied between $m = 48$ and $m = 44$ is 0.04 against 0.91 between $m = 44$ and $m = 12$; the deficits at the first pair are in the ratio 1.004 where a power fitted to the second requires 1.083. The measurement is exact, so that is not noise. A monotone dependence on m is not excluded — three points cannot exclude one — but no single power reaches them.

What the second cell shape measures, once the exponent is dropped, is **how tight Proposition 17's own error term is**. At depth 0 the deficit, the covariance and the pattern weight — in that order, which is not the table's — are 5.68, 4.36 and 1.95 per cent of $1/\ell = 1.05 \cdot 10^{-4}$, each first divided by $2C_2 E_c[A_S]$. That constant is the deficit's own factor in the four-term decomposition above, so dividing returns Δ_c there; the other two carry K and $2C_2$ instead, and are divided by the same constant only to put all three on one scale. Read straight off the table's entries the same three are 12.6, 9.7 and 4.3 per cent. Complete cancellation of the class-wise errors — what the diagonal class pairs alone would give, which the proof declines to claim — would put the deficit at 4.03 per cent of $1/\ell$ on that same scale. The proposition's error term carries an unspecified implied constant and a divisor sum above three, as the discussion after its proof records. So the asymptotic statement is not quantitative enough, at this finite band, to turn these percentages into a certified fraction of its bound: they are reported as a comparison with the $1/\ell$ that appears in the proof, and not as the fraction of the bound the measurement realises.

The measured deficit exceeds the diagonal-only figure at every multi-class cell, by 41, 30 and 16 per cent at $m = 48, 44, 12$, so the off-diagonal pairs add deficit rather than cancelling it. Those three are quoted and not summarised: an eight per cent change in m between the first two moves the excess by thirty-eight, so the excess is no more a function of m than the deficit is.

One caution belongs with those numbers, and it turns into a check. The registration fixed $D'_c = E_{\mathrm{same},c}[\mathfrak{S}_2] - 2$, and the rules are scored on that convention; but 2 is the *limiting* value E_∞, while (7) subtracts the band's own average $E_{\mathrm{all},B}$, which is 1.999986 here. The two differ by $1.39 \cdot 10^{-5}$, *larger than two of the four terms above*, so D'_c and D_c are two quantities under one name at this band size. Substituting the band's own average removes that same $1.39 \cdot 10^{-5}$ at every cell — half the departure from the main term at depth 0 and under two per cent of it at depth 3, since a constant offset cannot take a fixed share off departures spanning a factor of thirty. **T1 holds under both conventions**, so no verdict turns on the choice; the margin at depth 0 is $1.40 \cdot 10^{-5}$ against a shift of $1.39 \cdot 10^{-5}$, which is thin enough that another systematic of the same size would break it.

And the band's own departure decomposes into the same three terms, at a cell where the main term is known in advance to be zero. The band is all 105 classes modulo $2Q'$, so it is a cell of this partition — the one with $m = 105$ — and $K\,E^{\text{ex}}[A_S] - 2 = (175/128)(256/175) - 2 = 0$ exactly. Its $-1.38928 \cdot 10^{-5}$ is then $-2.37 \cdot 10^{-6}$ of pattern weight, $-5.83 \cdot 10^{-6}$ of divisor deficit and $-5.69 \cdot 10^{-6}$ of covariance, summing to the measured value to one part in 10^{10}. It is the only check here whose answer was fixed before the computation.

Because no exponent is claimed, *nothing here is to be applied to the first partition's row.* That row is the main term with no correction of any kind. A correction computed the way this measurement computes one would move its depth-4 entry, whose cell unions 34 classes; the entry is quoted uncorrected and the correction is not transported, since a law fitted on one partition and applied to another is exactly the transport this measurement was built to test rather than to assume.

The rule that tested the $1/m$ dilution is recorded neither held nor refuted, but *void*, and the reason is worth more than the rule. Its tolerance was 10^{-4}; the models it had to separate differ by 1.8 to $4.6 \cdot 10^{-5}$ at these m, so both pass it. A test the alternative also passes carries no evidence in either direction, and filing it as held would be worse than filing it as refuted, since a held rule is counted as support and there is none. The tolerance was set without asking what it had to discriminate; that question belongs before a rule is registered, not after it is scored.

Of the proof's errors it is the cell-sized one that is worth reading against the measurement, and this is where the sizes belong. The divisor-sampling term's *rate* does not vary with the depth: it is $O_\eta(B^\eta/\ell)$ with $\ell = B/2Q$, a band parameter, so one bound covers every cell of an octave. That the bound is uniform does not make the term itself constant across cells; what is fixed is the rate, and the constant in front of it is a separate question this does not settle.

No percentage can be read off it. The term carries an implied constant that the proof does not name, and the per-d error $O(1/\ell)$ enters summed against g, so the bound is $\ell^{-1}\sum_{d<B} g(d)$ and that divisor sum is not $O(1)$ — it grows like $\log B$, and at $B = 2 \cdot 10^6$ the single and double prime terms alone put it above three. Reading the term as $1/\ell$, which at that octave is one and a half per cent, is that bound with both the constant and the divisor sum set to 1, and it cannot then be compared with the two per cent that Measurement 20 registers as its cap: the proof supplies no cap for such a comparison to use.

What survives is the comparison's direction, and it survives a fortiori. The measurement caps the relative spread of D_c across the three octaves at two per cent and finds spreads from 0.0001 at depth 0 to 0.0128 at depth 5; the proof's bound on the error at each octave is unquantified, so it excludes nothing the measurement does not already exclude. **The proof's bound is weaker than the measurement at every cell.** It does not resolve what the exact spreads do, and nothing here should be read off it about which cell drifts and which does not. In particular it cannot be read against the sampled spreads to conclude that whatever moves the two refused depths is not an error term of this proof: the sampled spreads are the estimator's, the refusals with them, and the quantity's own spreads are the ones above.

Measurement 20 (the exponent, measured)**.** The exponent is discrete, and it is defined by ratios and not by a derivative: it is the one-step exponent $e_c(B)$ of (9), with $D_c(B)$ the value of (7) computed in the band of length B, the sign being fixed by the convention $|D_c| \sim B^{-e}$, so that a positive e means a decaying D_c. The e tabulated below is minus the least-squares slope of $\log|D_c|$ against $\log B$ over the three octaves. Their abscissae are the octave midpoints, spaced by exactly $\log 2$, so that quantity is the mean of the two one-step exponents; nothing is fitted that the ratios do not already determine.

Measured over $(10^6, 2\cdot 10^6]$, $(2\cdot 10^6, 4\cdot 10^6]$ and $(4\cdot 10^6, 8\cdot 10^6]$ at 400000 sampled pairs per cell, the exponents are

depth	0	1	2	3	4	5
e	+0.012633	+0.065546	+0.000936	+0.002904	+0.000241	−0.008428

— zero to three decimals at depth 4, within 0.003 at depths 2 and 3, and −0.008 at depth 5, against +0.013 and +0.066 at depths 0 and 1. The two large ones are where D_c is small: the spread across the three octaves runs from 0.0008 at depth 2 to 0.037 at depth 5, the last being the finite-cell drift of a cell of a few tens of elements, while D_c itself runs from 0.045 to 3.17, so the relative spread the exponent reads is large only where D_c is small. At ten times the sample the two move to −0.000879 and −0.016226, so depth 0 resolves to zero and depth 1 flips sign — the signature of noise rather than of a trend. That resampling is post hoc: it was run after M2 and M4 had been scored, and it does not lift either verdict. At the higher sample depth 1 still reads $e = -0.016226$ against M4's $|e| < 0.01$ and a spread of 0.0390 against M2's two per cent, so both stay refused as the code scores them.

The registration's own wording and the code's scoring of it differ, and this paper reports the stricter — the code's — throughout. Supplement S1 gives both readings and the verdicts each produces.

Sampling was not necessary. D_c is a deterministic functional of the cell, and the sum over all ordered pairs off the diagonal is one autocorrelation per cell: $\sum_{N \neq N' \in c} \mathfrak{S}_2(N - N') = \sum_{h \neq 0} \mathfrak{S}_2(h)\,(\mathbf{1}_c \star \mathbf{1}_c)(h)$, the diagonal being excluded on both sides because $\mathfrak{S}_2(0)$ is undefined; what is dropped is the term $(\mathbf{1}_c \star \mathbf{1}_c)(0) = n_c$. Computed that way the six read

depth	0	1	2	3	4	5
spread	0.0001	0.0004	0.0002	0.0002	0.0022	0.0128
e	−0.000059	−0.000285	−0.000136	−0.000164	−0.001584	−0.009202

— **six of six inside M2's two per cent and six of six inside M4's** $|e| < 0.01$, the shallow four by three orders. The two depths the registered rules refuse are the two where the sampling error is the largest fraction of D_c, and the refusals are a property of that estimator and not of D_c. The verdicts stand as scored, on the estimator the run registered. The deepest cell keeps the largest spread, 1.28 per cent, which Measurement 18 computes as 1.24 on its own denominator — the divisor deficit of a cell of a few tens of elements, and the only one of the six that the estimator was reading rather than adding to. Supplement S1 records the registered rules, what each was scored on, and where in the result files each refusal can be reproduced.

The sampling error of the pair average runs 0.001183 to 0.001370 at depths 0 to 4 — 0.001358 at depth 0, and lowest at depth 4, where the cell is smallest — and 0.000162 at depth 5, so at depth 1, where $D_c = 0.046766$ is six times smaller than any other, that error is the largest fraction of D_c anywhere in the table. That is what the refusals at depth 1 rest on, and the exact computation above says so in the other direction: remove the estimator and the spread there is 0.0004. Depth 0 is refused by the exponent rule alone and not by this: there the sampling error is 0.45% of D_c, and what fails is $e = +0.012633$ against $|e| < 0.01$, which the tenfold resampling resolves to −0.000879 and the exact computation to −0.000059. Depth 1 is quiet under the sign null ($z = +1.11$ in Measurement 10; in absolute value the third smallest of the six, behind depth 0's +0.1069 and depth 2's −0.2314), but that is a statement about the sign null: under the permutation null of Note 14, which this paper calls the stronger instrument, the same cell reads 47.71 — third of the six, behind the 106.89 and 102.22 at depths 3 and 4, and above the 42.42 at depth 5 that the paper calls the signal. The cell is therefore not one that carries no effect, and the two refusals are not excused by saying it does (`lab_cell_singular.py`, `lab_mask_placebo.py`).

Proposition 17 settles one thing and deliberately not another. Whatever produces the decay of the effect with N, it is not the arithmetic excess that produces the floor's size, since that excess does not decay to zero asymptotically. The decay exponent of the effect itself is not determined here. At the three shallowest depths no exponent is fitted, and the amplitude there is not uniformly below the floor across the range: at depth 1 it falls monotonically from +2.71 at the smallest octave to +0.69 at the largest, and two octaves cannot settle it unless they are the two smallest. The floor test asks of each scale separately whether the amplitude is present; whether it is falling is a question about the scales together, and failing the first does not answer the second.

6 Summary

- Lemma 2: the fluctuation of any cell mean under the sign null is exactly $Q_{cc}/n_c^2 - 2Q_{ca}/(n_c n) + Q_{aa}/n^2$; all three terms are needed, and they are computed from explicit finite correlations and weighted inner products.

- Observation 5 and Note 7: over the eight measured octaves that quantity is consistent with $(\log N)^{-1/2}$ at the shallow cells, where Note 3's fraction is constant, and is inconsistent with count-based $n_c^{-1/2}$ scaling over the same range. The exponent itself is measured at all six depths and the deepest fits 0.107306, which the form does not predict. A count-based interval is narrower than the exact Rademacher-null scale by a factor 9.3 across a factor 128 in N, and absolutely by factors of 7.3 to 158 at the top octave, the largest of them at the shallowest cell.

- Lemma 8 and Measurement 10: neither the cell effect nor the floor is reproduced under label permutation, so both read the correspondence and not the cell sizes — the floor least expectedly, since one expects a floor to be a property of the sizes. The floor is a second measurement of the same correspondence. What it collapses to is the count-based bar (Lemma 12), so the two floors are two questions and not a bracket; z_c is quoted against the sign null throughout.

- Proposition 17: what is established is an asymptotic statement, not an identity. D_c equals a cell-determined constant up to $O(1/\ell)$ and $O_\eta(B^{\eta-1})$, both of which vanish with the band and neither of which carries a named implied constant; so no rate and no percentage may be read from it, and its consequence — that this quantity is not what makes the effect decay — holds in the limit and not at any fixed band. The argument is given in §5. It is the one labelled statement here that has had neither a machine check nor a reading by a subject expert, and its hardest step is named as such in its own proof. Two of the six depths are refused by the registered rules, and both refusals belong to the sampling estimator rather than to D_c: the same measurement computes D_c exhaustively, by one autocorrelation per cell, and done that way all six depths sit inside both registered caps, the shallow four by three orders. The verdicts stand as scored, on the estimator the run registered; Supplement S1 gives them. It is the proposition's own error term, not a failure of it.

- Measurement 18: that quantity's main term is a finite rational over the residue classes the cells are unions of, it is nonzero at every depth — which is the hypothesis Proposition 17 needs and does not otherwise have — and multiplied by a constant with no fitted parameter it reproduces the measured profile, four of six inside two sampling errors across cells spanning a factor of 68. **It is the only quantity here computed from the arithmetic alone**: Lemma 2 also gives a size in closed form, but from the field's own values, and this takes no input from the field at all. It is post hoc in timing. A registered rule of it was refuted: it named the worst relative deviation at the wrong cell, having been written on the unstandardised statistic while the standardisation was printed beside it.

- Two things are needed for the floor, and only the second is what the placebo removes: coherence of $u_c(v)$ across the cell, and a cell correlated with the arithmetic that coherence comes from. Nothing here shows that pair is special to this field, and nothing here shows it is not. Sharing the arithmetic input is not by itself enough: Lemma 12 gives a random cell of the same band a floor whose permutation average is exactly the count-based one, and that cell shares the input.

All preregistered checks, including the failed ones and the one recorded void, are reported with their verdicts in Supplement S1 (Appendix B). Several numerical patterns conjectured in the course of the work were rejected there; none is used as an assumption in anything stated above.

References

[1] J.-J. Huang and H. Li, *On the connection between the Goldbach conjecture and the Elliott–Halberstam conjecture*, arXiv:2005.03811 [math.NT]; v1 (May 2020), v2 (August 2022). Also published in a Springer volume, `doi:10.1007/978-3-030-67996-5_17`. References here are to the arXiv version, which we have read, and not to the published version, which we have not.

[2] L. Mirsky, *The number of representations of an integer as the sum of a prime and a k-free integer*, Amer. Math. Monthly **56** (1949), 17–19. Theorem 1, taken at $k = 2$: the count is $\prod_{p \nmid n}(1 - p^{1-k}/(p-1)) \operatorname{Li} n + O(n/\log^H n)$ for every $H > 0$, whose local factor at $k = 2$ is the $1 - 1/(p(p-1))$ used above.

[3] P. X. Gallagher, *On the distribution of primes in short intervals*, Mathematika **23** (1976), 4–9; corrigendum, Mathematika **28** (1981), 86. What is used here is the theorem on the average of the singular series over k-tuples. We have not obtained the original; the statement is used in the form in which [4] states and reproves it.

[4] J. Pintz, *On the singular series in the prime k-tuple conjecture*, arXiv:1004.1084v1 [math.NT], 2010. What is used above is Theorem 1, $\mathfrak{S}_{\mathcal{H}}(H) = 1 + O(\varepsilon)$ for $H \geq \exp(k^{1/\varepsilon})$, applied with $\mathcal{H}$ a single element: there $\mathfrak{S}(\mathcal{H}) = 1$ and $\mathfrak{S}(\mathcal{H} \cup \{h\})$ is the singular series of the shift, so the average over shifts tends to 1. Equation (1.5), $\sum_{|H|=k,\, H\subset[1,H]} \mathfrak{S}(H) \sim H^k$, is Gallagher's global average and is *not* what is used: its $k = 1$ case is the sum over singletons, and a singleton has $\mathfrak{S} = 1$ identically, so that case is an identity carrying no information about shifts. The paper itself draws the distinction, noting that $\mathfrak{S}_{\mathcal{H}}(H) \to 1$ "yields more information about the singular series than the global average". §1 also records $\mathfrak{S}(\mathcal{H} \cup \{h\}) = 0$ for odd h, which is the vanishing that makes the average over even shifts twice the average over all.

[5] W. G. Cochran, *Sampling Techniques*, 3rd ed., Wiley, 1977. (Finite-population correction for the variance of a sample mean without replacement.)

[6] L. Kish, *Survey Sampling*, Wiley, 1965. (Design effect and effective sample size for correlated observations.)

[7] E. L. Lehmann and J. P. Romano, *Testing Statistical Hypotheses*, 3rd ed., Springer, 2005. (Permutation and randomisation tests conditional on the observed configuration.)

[8] A. Wintner, *Random factorizations and Riemann's hypothesis*, Duke Math. J. **11** (1944), 267–275, `doi:10.1215/S0012-7094-44-01122-1`.

Disclosure statement

No potential competing interest, financial or non-financial, was reported by the author. No funding was received for this work. An AI assistant was used in preparing it; what tool, how, and why is set out in Appendix B, together with what was checked by machine, what by computation, and what by neither.

Data availability statement

The code behind every figure printed here, the output it produced, and the Lean 4 development are openly available at `https://github.com/soke91/arithmetic-convolution-fields`, in the directory `P4-coherent-cell-floor-r1/`. The figures quoted are pinned by the SHA-256 list in that directory, as Appendix B sets out. No data derived from human or animal subjects were used; every quantity here is computed from the integers.

A The scale of the normaliser

Observation 5's heuristic reads the scale $N \log N$ off the asymptotic size of $V(N)$. That asymptotic is stated and proved here so that this paper depends on no unpublished manuscript; nothing else in the paper uses it, every computation dividing by $V(N)$ as computed.

Proposition 21 (the scale of V)**.** *Let* $V(N) = \sum_{v<N} \mu^2(v)\Lambda(N-v)^2$ *and* $W(N) = \sum_{w<N} \Lambda(w)^2$*, and put* $\mathfrak{A}(N) = \prod_{q\nmid N}\big(1 - 1/(q(q-1))\big)$*. Then* $V(N) = W(N)\mathfrak{A}(N)(1+o(1))$*, and hence* $V(N) \sim \mathfrak{A}(N)\, N \log N$*, the second form using* $W(N) \sim N \log N$*.*

Proof. $\Lambda(w)^2$ is supported on prime powers with weight $(\log p)^2$, so with $w = N - v$,

$$V(N) = \sum_{p^k<N} (\log p)^2 \mu^2(N-p^k) = \sum_{p<N} (\log p)^2 \mu^2(N-p) + O(\sqrt{N}\log^2 N),$$

a $(\log p)^2$-weighted count of squarefree shifted primes. Its local density at a prime q is 1 when $q \mid N$ — there $q^2 \mid N-p$ forces $q \mid p$, hence $p = q$, a single term — and $1 - 1/\varphi(q^2) = 1 - 1/(q(q-1))$ when $q \nmid N$, the bad class $p \equiv N \pmod{q^2}$ being a unit class. The count is then Mirsky's theorem on squarefree values of shifted primes [2], a statement at the endpoint $x = N$. The weight needs no uniform-in-x partial summation, but the cut has to move: for *fixed* ε the ratio $(\log p)^2/(\log N)^2$ on $p > N^{1-\varepsilon}$ lies in $((1-\varepsilon)^2, 1]$ and does not tend to 1, so that choice gives $(1+O(\varepsilon))$ and not $(1+o(1))$. Take instead $\varepsilon = \varepsilon(N) \to 0$ slowly enough that $\varepsilon \log N/\log\log N \to \infty$ — $\varepsilon = 1/\log\log N$ serves. Then $(1-\varepsilon)^2 = 1 + o(1)$, so $(\log p)^2 = (1+o(1))(\log N)^2$ on $p > N^{1-\varepsilon}$, while $\pi(N^{1-\varepsilon}) = o(N/\log N)$ still absorbs the smaller primes whatever their weight, so the weight is $(1+o(1))(\log N)^2$ on all but $o(N/\log N)$ of the terms. The second form follows from $W(N) \sim N\log N$, which is partial summation from $\theta(x) \sim x$ and is the only analytic input either form takes. $\square$

The local factor here is $\mathfrak{A}$ and not the Hardy–Littlewood singular series $\mathfrak{S}$: both depend on N only through the primes dividing it, but $\mathfrak{S}$ through a product *over* those primes and $\mathfrak{A}$ through a product over the primes *not* among them, so a substitution of one for the other is easy to make and hard to see.

B Reproducibility

The code, the output it produced, and the Lean development are at

`https://github.com/soke91/arithmetic-convolution-fields` in the directory `P4-coherent-cell-floor-r1/`, which holds this revision. The directory `P4-coherent-cell-floor/` beside it holds the version sent to the journal and is frozen, so a reader comparing the two is reading a revision against what was actually submitted rather than against a working copy. That address is mutable, and this revision is not in a tagged release: the tagged `v1.0.1`, commit `b2f365d`, predates the revision directory, and its second-cell scripts and outputs are not the ones read here. What pins the bytes travels with them instead — `PACKET.json` in that directory carries a SHA-256 for each of the thirty files in `code/` and `results/`, so a figure can be checked against the exact source that produced it without resolving a tag. The pin is on those and not on this document, which cannot cite a hash that covers it. Each entry in the table below is run as `python code/<script>.py` from that directory, and each writes the file of the same name in `results/`. The scripts import only the standard library and NumPy; they never import one another. Five of them do read a result file that another script produces, so the packet carries those producers too and the order matters: `lab_cell_singular` reads `lab_cell_floor` and `lab_mask_placebo`; `lab_secondcell2_predict` reads `lab_secondcell` and `lab_secondcell_predict`; `lab_secondcell2` reads both `predict` files; `lab_maincoef` reads `lab_cell_singular` and `lab_secondcell2`; and `audit_cn_coin_deep` reads `audit_cn_class` and `audit_cn_multnull_deep`. Run in that order, every input exists before the script that reads it.

The figures printed here were produced under Python 3.12.10 and NumPy 2.2.5. `lab_maincoef.py` and the two `predict` files compute in exact rational arithmetic and their output is reproducible bit for bit on any platform; the rest compute in double precision, where a different BLAS may move the last printed digit of a summed quantity. No figure quoted in this paper is stated to more digits than that distinction supports.

Four of the runs exit with a nonzero status. That is their registered behaviour and not a failure: a run whose own preregistered rule was refuted says so in its exit code as well as in its output, and the refutations are reported in Supplement S1.

The Lean development is in `P4-coherent-cell-floor-r1/lean/`, with Mathlib pinned in `lake-manifest.json` and the toolchain in `lean-toolchain`; `lake build` reproduces `axioms.txt`, which is the axiom report quoted above.

Script	Result file	Supports
`lab_cellmom_montecarlo.py`	`lab_cellmom_montecarlo.txt`	Measurement 4
`lab_cell_floor.py`	`lab_cell_floor.txt`	Measurement 6
`lab_mask_placebo.py`	`lab_mask_placebo.txt`	Measurement 10
`lab_cell_singular.py`	`lab_cell_singular.txt`	Measurements 20, 15
`lab_maincoef.py`	`lab_maincoef.txt`	Measurement 18
`lab_secondcell2.py`	`lab_secondcell2.txt`	Measurement 19
`audit_cellfloor_sdz.py`	`audit_cellfloor_sdz.txt`	Note 7
`audit_cn_coin_deep.py`	`audit_cn_coin_deep.txt`	Note 11's 0.6455

Use of an AI assistant, and what checked what

This work was done in close collaboration with an AI assistant, which carried much of the derivation and all of the computation. The author takes full responsibility for every claim made here.

Which tool, and how. The manuscript in its present form was prepared with Anthropic's Claude, model Opus 5, run through the Claude Code command-line interface. It was used to derive and check the algebraic identities, to write and run every script in the accompanying code, to write the Lean development, and to draft and revise this text. Earlier stages of the work, before the material reported here took its present shape, used earlier models of the same family; the repository does not record which, and this paper does not rely on anything from those stages that has not since been rederived and rerun. The reason for use is throughput on a body of computation and case analysis larger than the author would otherwise have carried out, and the cost of that choice is the one this note is about: what a reader can check does not depend on trusting the tool.

What follows says what was checked and by what, rather than asserting that it was checked.

Machine-verified. Lemma 2's algebraic identity, Lemma 12's closed form and Lemma 8 are proved in Lean 4 against Mathlib; the axiom report is in the code packet and each depends on `propext`, `Classical.choice` and `Quot.sound` and on nothing else. **Proposition 17 is not machine-verified and has had no reading by a subject expert.** Its statement is written out in Lean and left unproved, so the axiom report carries `sorryAx` on that line and the absence is checkable rather than a matter of trust. The finite half of its argument is verified separately; the analytic half, the Euler product and its convergence, is not.

Checked by computation. Every figure quoted is produced by a named run in the accompanying code, and a check in the build refuses a figure that no cited run prints. Every registered rule is reported with its verdict, including the thirteen the ten named runs did not meet, the fourteenth recorded in a separate output, and the one recorded void. The packet preserves the second-cell prediction and measurement artifacts, but does not make a claim about their historical ordering.

Supplement S1: what rests on what

Lemma 1, Lemma 2, Lemma 8, Lemma 12, Lemma 16, Proposition 17 and Proposition 21 are proved and use no computation; those seven are every proof in the paper. Lemmas 1 and 16 use no arithmetic either: the first is a quadratic form in an arbitrary covariance, the second a summation bound for an arbitrary modulus, and neither is among the statements formalised in Lean. The scripts were written against the manuscript of the generation they belong to and their headings name its statements, not this paper's: `prop:coh`, `prop:placebo`, `rem:floorsignal` and `rem:sdzrange` are Observation 5, Lemma 8, Note 11 and Note 7 under earlier names, and `lem:coin` is the sign null, as Note 14 records. `rem:copiedfloor`, `rem:levelmeas`, `rem:mcratios` and `rem:cncoindeep` are four further statements of that manuscript which this paper does not carry; the runs are cited here for their numbers and nothing above rests on those four. The Supports column names one statement per run and there are more. `lab_cell_floor.py` also carries Measurement 9, the paragraph of Measurement 4 that reads Note 3's structural constant at the top octave — the one that measurement itself ends by attributing to this file — the table of Observation 5, Notes 3 and 7, and two items of §1.3 — the one on the shallow depths not being uniformly quiet, and the one on Observation 5's looseness factors, whose two inputs are that file's Q_{cc}/n_c^2 row and its heuristic 0.049540. `lab_mask_placebo.py` also carries Notes 11, 13 and 14 and the se_c row of Measurement 15. The width ratio 0.6455 inside Note 11 is not that file's: it comes from `audit_cn_coin_deep.py`, the last run of the table above. Counting a named statement as resting on a run when it prints a figure that only that run produces, five rest on the first file — Observation 5, Note 3 and Measurements 4, 6 and 9 — and six on the second: Measurement 10, Notes 11, 13 and 14, the se_c row of Measurement 15, and the sign-null amplitudes that Measurement 20 reads off it. The unit is stated because the count turns on it: counting the items of §1.3 and the paragraphs that carry no figure of their own alongside the named statements would make the first eight. Under a unit that can be checked the two numbers are five and six. The measurements state the band, the cell index and the statistic they were taken of, and each was run with its decision rule fixed in advance. Two qualifications belong beside that sentence. The first is that not every registered rule can fail. N3 of Measurement 4 reads the Monte-Carlo precision at the smaller draw count against the figure an earlier printing quoted; the precision is a function of the draw count alone, and the draw count is a constant of the script, so N3 evaluates the same arithmetic on every run and cannot be refuted by any input. It is an identity, and it is counted among the registered rules below. The second is that one registration block carries text written after its run. The heading of `lab_mask_placebo.py` reads "written before this script was run", and L5 inside it argues for its own cap by citing the factors 3.8 to 105 that the run produced, while L4 records in the same block what the run failed to test. The rules themselves were fixed first — what came later is the reasoning printed beside them — but a reader who takes the heading literally is reading two layers as one. One quantity a measurement leans on carries no rule of its own: the spread of se_c^2/D_c that Measurement 15 calls settling — 25.4% as printed, 19.8% under the matched convention. The 19.1% that the cited file also prints is the diagonal-out row, which is an over-correction and which that measurement withdraws. What is registered there is M3, the correlation; the ratio came out of the same run and is gated by nothing. The enumeration over all 720 relabellings that scores M3 *is* registered — that file declares it in its heading — and the ten-fold resampling of Measurement 20 is post hoc and says so. The permutation control of Lemma 8 was *not* run against each cell-indexed claim. It was run once: ten label permutations on the single octave $(2 \cdot 10^6, 4 \cdot 10^6]$, with the cells indexed by depth. Of the six cell-indexed measurements the rest carry no control by *that* permutation — Measurement 15 carries its own, an enumeration over all 720 orderings of the depth labels — and two of the cited files say so where they say it best — one records that the placebo "is not run here", and the other that the control the placebo file supplies is not the control that statement needs, since it permutes labels for a different statistic. The control is not claimed for the others, and what the evidence files say is what stands.

Supplement S1: the registered rules and their verdicts

Six of the ten named runs record a rule their own output did not meet. The separately retained first second-cell output also records P1 as refuted. The unit of the count is the pair (run, label), because the labels are local to their files and repeat across them: `N1` names one rule in `lab_cellmom_montecarlo` and a different one in `lab_maincoef`, and `F1` and `F2` likewise name different rules in `lab_secondcell_predict` and `audit_cn_coin_deep`. Under that unit thirty-eight rules carry a verdict — twenty-four hold, thirteen refuted, one void — and one further label, W3, sits under the same registration heading as W1, W2 and W4 and is reported without a verdict, so thirty-nine labels are registered in all. The fourteen are named here because a reader who does not open the files would otherwise take the runs for clean, and because in every case what the rule asked for is not what the statement above says. The seventh, `lab_cellmom_montecarlo`, met all four of its rules, and two of those four score a 60-draw quotation that an earlier printing of this work made and this paper does not: the table above is at 2000 draws, and the precision Measurement 4 reasons from is computed in the text rather than gated by any rule of that file. The ratios that measurement states are checked in a block the file marks post hoc.

`lab_cell_floor` registered that the fitted exponent b in $\mathrm{se} \sim N^{-b}$ over the eight octaves, and the quantity $1/(2\langle \log N \rangle)$ beside it, would match the values published earlier (C4); that the row of Q_{cc}/n_c^2 would have minimum 0.124 and maximum 0.307 at the top octave (C3); and that the four ratios of C1 would behave the same way at the octave $(2 \cdot 10^6, 4 \cdot 10^6]$ (C2). C4 is refuted by one line of four: the three fitted b agree with what was published, and $1/(2\langle \log N \rangle)$ comes out 0.036038 against a published 0.0358 — the value used in this paper is the new one, and nothing above turns on the third decimal. C3 is refuted on its minimum, which measures 0.121208 against the registered 0.124: the row's minimum sits at depth 1 rather than at depth 0, so the published figure was reading the wrong end. Two of C3's three clauses fail, not one: the maximum holds at 0.307074 against the registered 0.307, but the heuristic reads 0.049540 against 0.049, which is 0.000540 away and so outside the 0.0005 the rule allowed — it fails as registered, by a margin smaller than the last digit of the published value it was registered against. C2 is refuted at depth 5 alone, which reads 0.557654 at that octave against 0.551516 at the top; that is the sentence in Measurement 4 which says depth 5 does not. `lab_cell_singular` registered four things and three fail: that D_c is positive and increasing in the depth (M1); that it is invariant across octaves — a spread within two per cent (M2) and a fitted exponent with $|e| < 0.01$ (M4), a threshold its result file does not print; and that the floor tracks D_c across depths, a correlation above 0.8 (M3), which holds at 0.9805 and is the control Measurement 15 rests on. Only M1 is about the depth, and its failure is the sentence in Measurement 15 that neither row is monotone; M2 and M4 are about scale and fail only where D_c is small, as recorded above. `lab_mask_placebo` registered that the floor would survive permutation within a tenth (L3) — it does not, the collapse being the subject of Measurement 10 and the reason Note 11 and Lemma 12 were written — and two things at once as L4: that the true labelling's z_c is monotone decreasing in depth, and that under permutation the rank correlation between depth and z_c has $|\text{mean}| < 0.3$. L4 is refuted on the first — z rises by 1.0065 from depth 0 to depth 1 — while the second holds, the permuted mean Spearman being -0.0114 against a true -0.9429. `audit_cellfloor_sdz` registered that its printed pair would be reproduced under one of two readings (W1) and that the ratio would grow with the cell (W2); neither does, and the convention that settles W1 is stated in Note 7. `audit_cn_coin_deep` registered four things about the two sign ensembles and two fail: F3 asked the i.i.d. ensemble to cover the true value at the deepest cell and no draw of it does, and F4 asked the two nulls to agree in width within a factor 1.5 and they do not — the 0.6455 that Note 11 quotes is that disagreement. F1 and F2 hold. This run is cited here for the width ratio alone, and its two refutations are the reason that ratio is worth quoting.

Seven of the fourteen concern a prediction about a row this paper prints — its shape across depths (C3, M1, L4, W2) or its constancy across octaves (C2, M2, M4) — and the statements above report what the rows do. Two more, F3 and F4, are about the sign null's own reach rather than about this field, and are named above. A tenth, N3, registered which of six cells carries the worst deviation and named depth 5; its own file records the refutation together with the reason, that the rule was registered on an unstandardised quantity and so asked about a different question from the one it appears to ask. P1 is the omitted fourteenth: the first second-cell run scored one sampled draw at three depths, and its depth-2 draw missed the registered three-standard-error cap. Its result file records it as refuted; it is the separate output noted above, not one of the ten named runs. Nothing above rests on that sampled rule. The other three did more than that, and are listed here because the count alone would hide them. W1 registered that the pair (5.8, 160) would be reproduced under one of two readings; it was not, and the pair this paper prints, in Note 7 and in the Summary, is (7.3, 158), the measured one. C4 likewise changed a number this paper prints: $1/(2\langle \log N \rangle)$ measures 0.036038 against the published 0.0358, and it is the measured value that appears in Measurement 6 and in Note 7. L3 registered no shape at all but a size — that the floor would survive permutation within a tenth — and its refutation is what §4 was written around: Note 11, Lemma 12 and Note 13 all follow from it.